Detailed Derivations for the

"Comment on 'Casimir Energy in Kerr Space-time'"

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November 6, 2026

Version 1.0.0

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I. Four-Velocity, Normalization Factor, and Comoving Coordinates

In the coordinate system (t, r, θ, ϕ) (the order of θ, ϕ is reversed relative to that in the paper), the four-velocity is

$$w^\mu = \left(\frac{dt}{d\tau}, \frac{dr}{d\tau}, \frac{d\theta}{d\tau}, \frac{d\phi}{d\tau}\right)$$

On the equatorial plane, $\theta = \frac{\pi}{2}$, $\frac{d\theta}{d\tau} = 0$, and for a circular orbit, $r = r_0$, $\frac{dr}{d\tau} = 0$. Hence,

$$w^\mu = \left(\frac{dt}{d\tau}, 0, 0, \frac{d\phi}{d\tau}\right)$$

The angular velocity $\Omega = \frac{d\phi}{dt}$ is the angular velocity measured by an observer at infinity, using coordinate time. Thus,

$$\begin{aligned}\frac{d\phi}{d\tau} &= \frac{d\phi}{dt} \cdot \frac{dt}{d\tau} \\ &= \Omega \cdot \frac{dt}{d\tau}\end{aligned}$$

We obtain

$$w^\mu = \frac{dt}{d\tau}(1, 0, 0, \Omega)$$

$\delta^\mu_t = (1, 0, 0, 0)$ and $\delta^\mu_\phi = (0, 0, 0, 1)$, where δ is the Kronecker delta. Hence,

$$w^\mu = \frac{dt}{d\tau}\left(\delta^\mu_t + \Omega\delta^\mu_\phi\right)$$

The normalization condition is $w^\mu w_\mu = 1$. (The paper uses the metric signature $(+, -, -, -)$, which is opposite to the convention commonly used in general relativity; the normalization condition is therefore not $w^\mu w_\mu = -1$.)

$$g_{\mu\rho} w^\rho = w_\mu$$

$$\begin{aligned}w_t &= g_{tt}w^t + g_{tr}w^r + g_{t\theta}w^\theta + g_{t\phi}w^\phi \\ &= g_{tt}\frac{dt}{d\tau} + 0 + 0 + \Omega g_{t\phi}\frac{dt}{d\tau} \\ &= \frac{dt}{d\tau}(g_{tt} + \Omega g_{t\phi})\end{aligned}$$

$$\begin{aligned}w_r &= g_{rt}w^t + g_{rr}w^r + g_{r\theta}w^\theta + g_{r\phi}w^\phi \\ &= \frac{dt}{d\tau}(g_{rt} + \Omega g_{r\phi})\end{aligned}$$

$$w_\theta = \frac{dt}{d\tau}(g_{\theta t} + \Omega g_{\theta\phi})$$

$$w_\phi = \frac{dt}{d\tau}(g_{\phi t} + \Omega g_{\phi\phi})$$

$$\begin{aligned}w^\mu w_\mu &= w^t w_t + w^r w_r + w^\theta w_\theta + w^\phi w_\phi \\ &= \frac{dt}{d\tau}\cdot\frac{dt}{d\tau}(g_{tt} + \Omega g_{t\phi}) + 0 + 0 + \Omega\frac{dt}{d\tau}\cdot\frac{dt}{d\tau}(g_{\phi t} + \Omega g_{\phi\phi}) \\ &= \frac{dt}{d\tau}\cdot\frac{dt}{d\tau}\left[g_{tt} + \Omega g_{t\phi} + \Omega\left(g_{\phi t} + \Omega g_{\phi\phi}\right)\right] \\ &= \left(\frac{dt}{d\tau}\right)^2\left(g_{tt} + \Omega g_{t\phi} + \Omega g_{\phi t} + \Omega^2 g_{\phi\phi}\right)\end{aligned}$$

$$\left(\frac{dt}{d\tau}\right)^2\left(g_{tt}+\Omega g_{t\phi}+\Omega g_{\phi t}+\Omega^2 g_{\phi\phi}\right)=1$$

$$\Rightarrow \frac{dt}{d\tau}=\sqrt{\frac{1}{g_{tt}+\Omega g_{t\phi}+\Omega g_{\phi t}+\Omega^2 g_{\phi\phi}}}$$
$$=\left(g_{tt}+\Omega g_{t\phi}+\Omega g_{\phi t}+\Omega^2 g_{\phi\phi}\right)^{-1/2}$$
$$=\left(g_{tt}+2\Omega g_{t\phi}+\Omega^2 g_{\phi\phi}\right)^{-1/2}$$

Here, $g_{\phi t}=g_{t\phi}$. Defining $C\left(\Omega\right):=\left(g_{tt}+2\Omega g_{t\phi}+\Omega^2 g_{\phi\phi}\right)^{-1/2}=dt/d\tau$, we have

$$w^\mu=\frac{dt}{d\tau}\left(\delta_t^\mu+\Omega\delta_\phi^\mu\right)$$
$$=\left(g_{tt}+2\Omega g_{t\phi}+\Omega^2 g_{\phi\phi}\right)^{-1/2}\left(\delta_t^\mu+\Omega\delta_\phi^\mu\right)$$
$$=C\left(\Omega\right)\left(\delta_t^\mu+\Omega\delta_\phi^\mu\right)$$

Solving $g_{tt}+2\Omega g_{t\phi}+\Omega^2 g_{\phi\phi}=0$ (using $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$ for the quadratic equation $ax^2+bx+c=0$) gives

$$\Omega_\pm^{'}=\frac{-2g_{t\phi}\pm\sqrt{4g_{t\phi}^2-4g_{\phi\phi}g_{tt}}}{2g_{\phi\phi}}$$
$$=-\frac{g_{t\phi}}{g_{\phi\phi}}\pm\frac{\sqrt{g_{t\phi}^2-g_{\phi\phi}g_{tt}}}{g_{\phi\phi}}$$

The line element is:

$$ds^2=\left(1-\frac{2Mr}{\Sigma}\right)dt^2+2\frac{2Mar}{\Sigma}\sin^2\theta dtd\phi-\frac{\Sigma}{\Delta}dr^2-\Sigma d\theta^2-\frac{A}{\Sigma}\sin^2\theta d\phi^2$$
$$=g_{tt}dt^2+2g_{t\phi}dtd\phi+g_{rr}dr^2+g_{\theta\theta}d\theta^2+g_{\phi\phi}d\phi^2$$

where

$$\Sigma=r^2+a^2\cos^2\theta,$$
$$\Delta=r^2+a^2-2Mr,$$
$$A=\left(r^2+a^2\right)^2-a^2\Delta\sin^2\theta$$
$$=\left(r^2+a^2\right)\Sigma+2Mra^2\sin^2\theta,$$
$$a=\frac{J}{M}.$$

The metric matrix is

$$g_{\mu\nu}=\begin{pmatrix}g_{tt}&g_{tr}&g_{t\theta}&g_{t\phi}\\g_{rt}&g_{rr}&g_{r\theta}&g_{r\phi}\\g_{\theta t}&g_{\theta r}&g_{\theta\theta}&g_{\theta\phi}\\g_{\phi t}&g_{\phi r}&g_{\phi\theta}&g_{\phi\phi}\end{pmatrix}$$

Here, $g_{t\phi}=g_{\phi t}=\frac{2Mar}{\Sigma}\sin^2\theta$, and

$$g_{t\phi}dtd\phi+g_{\phi t}d\phi dt=\frac{2Mar}{\Sigma}\sin^2\theta dtd\phi+\frac{2Mar}{\Sigma}\sin^2\theta d\phi dt$$
$$=\frac{4Mar}{\Sigma}\sin^2\theta dtd\phi$$

The coordinate transformation is:

$$\begin{cases}t'=t\\r'=r\\\theta'=\theta\\\phi'=\phi-\Omega t\end{cases}$$

The inverse transformation is:

$$\begin{cases} t = t' \\ r = r' \\ \theta = \theta' \\ \phi = \phi' + \Omega t' \end{cases}$$

This gives

$$\begin{aligned} ds^2 &= g_{tt}dt^2 + 2g_{t\phi}dtd\phi + g_{rr}dr^2 + g_{\theta\theta}d\theta^2 + g_{\phi\phi}d\phi^2 \\ &= g_{tt}dt'^2 + 2g_{t\phi}dt'd\left(\phi^{'} + \Omega t'\right) + g_{rr}dr'^2 + g_{\theta\theta}d\theta'^2 + g_{\phi\phi}d\left(\phi^{'} + \Omega t'\right)d\left(\phi^{'} + \Omega t'\right) \\ &= g_{tt}dt'^2 + 2g_{t\phi}dt'\left(d\phi^{'} + \Omega dt'\right) + g_{rr}dr'^2 + g_{\theta\theta}d\theta'^2 + g_{\phi\phi}\left(d\phi^{'} + \Omega dt'\right)\left(d\phi^{'} + \Omega dt'\right) \\ &= g_{tt}dt'^2 + 2g_{t\phi}\left(dt'd\phi^{'} + \Omega dt'^2\right) + g_{rr}dr'^2 + g_{\theta\theta}d\theta'^2 + g_{\phi\phi}\left(d\phi^{'2} + \Omega d\phi^{'}dt' + \Omega dt'd\phi^{'} + \Omega^2 dt'^2\right) \\ &= g_{tt}dt'^2 + 2g_{t\phi}\left(dt'd\phi^{'} + \Omega dt'^2\right) + g_{rr}dr'^2 + g_{\theta\theta}d\theta'^2 + g_{\phi\phi}\left(d\phi^{'2} + 2\Omega dt'd\phi^{'} + \Omega^2 dt'^2\right) \\ &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right)dt'^2 + 2\left(g_{t\phi} + \Omega g_{\phi\phi}\right)dt'd\phi^{'} + g_{rr}dr'^2 + g_{\theta\theta}d\theta'^2 + g_{\phi\phi}d\phi^{'2} \\ &= g^{'}_{\mu\nu}dx^{'\mu}dx^{'\nu} \end{aligned}$$

$$\begin{cases} g^{'}_{tt} = \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) \\ g^{'}_{t\phi} = g^{'}_{\phi t} = \left(g_{t\phi} + \Omega g_{\phi\phi}\right) \\ g^{'}_{rr} = g_{rr} \\ g^{'}_{\theta\theta} = g_{\theta\theta} \\ g^{'}_{\phi\phi} = g_{\phi\phi} \end{cases}$$

When the angular velocity Ω of an observer moving along a circular equatorial orbit satisfies $g_{t\phi} + \Omega g_{\phi\phi} = 0$, or equivalently $\Omega = -\frac{g_{t\phi}}{g_{\phi\phi}}$, we have $2\left(g_{t\phi} + \Omega g_{\phi\phi}\right)dt'd\phi^{'} = 0$. The $dt'd\phi^{'}$ cross term is eliminated, and there is no frame-dragging effect in this frame (the local reference frame of the observer on the circular equatorial orbit). Such an observer is also called a zero-angular-momentum observer (ZAMO). We define this angular velocity as

$$\omega_d = -\frac{g_{t\phi}}{g_{\phi\phi}}$$

$$\begin{aligned} \omega_d &= -\frac{g_{t\phi}}{g_{\phi\phi}} \\ &= -\frac{\frac{2Mar}{\Sigma}\sin^2\theta}{-\frac{A}{\Sigma}\sin^2\theta} \\ &= \frac{2Mar}{A} \end{aligned}$$

This corresponds to the first term in $\Omega_{\pm}$. For the second term, on an equatorial orbit $\theta = \pi/2$, with the metric components treated as constants (see the detailed analysis below), we have

$$\begin{aligned} \left.\frac{\sqrt{g_{t\phi}^2 - g_{\phi\phi}g_{tt}}}{g_{\phi\phi}}\right|_{\theta=\pi/2} &= \left.\frac{\sqrt{\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2 - \left(-\frac{A}{\Sigma}\sin^2\theta\right)\left(1 - \frac{2Mr}{\Sigma}\right)}}{-\frac{A}{\Sigma}\sin^2\theta}\right|_{\theta=\pi/2} \\ &= -\frac{\Sigma\sqrt{\Delta}}{A} \end{aligned}$$

where

$$\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2-\left(-\frac{A}{\Sigma}\sin^2\theta\right)\left(1-\frac{2Mr}{\Sigma}\right)\Big|_{\theta=\pi/2}=\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2-\left(-\frac{A}{\Sigma}\sin^2\theta\right)\left(1-\frac{2Mr}{\Sigma}\right)$$
$$=\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2+\frac{1}{\Sigma}\left[\left(r^2+a^2\right)\Sigma+2Mra^2\sin^2\theta\right]\sin^2\theta\left(1-\frac{2Mr}{\Sigma}\right)$$
$$=\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2+\left[\left(r^2+a^2\right)+\frac{2Mra^2}{\Sigma}\sin^2\theta\right]\sin^2\theta\left(1-\frac{2Mr}{\Sigma}\right)$$
$$=\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2+\left(r^2+a^2\right)\sin^2\theta\left(1-\frac{2Mr}{\Sigma}\right)+\left[\frac{2Mra^2}{\Sigma}\left(\sin^2\theta\right)^2-\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2\right]$$
$$=\left(r^2+a^2\right)\sin^2\theta\left(1-\frac{2Mr}{\Sigma}\right)+\frac{2Mra^2}{\Sigma}\left(\sin^2\theta\right)^2$$
$$=\left(r^2+a^2\right)\sin^2\theta-\left(r^2+a^2\right)\sin^2\theta\cdot\frac{2Mr}{\Sigma}+\frac{2Mra^2}{\Sigma}\left(\sin^2\theta\right)^2$$
$$=\left[\left(r^2+a^2\right)\sin^2\theta-r^2\frac{2Mr}{\Sigma}\sin^2\theta-\frac{2Mra^2}{\Sigma}\sin^2\theta+\frac{2Mra^2}{\Sigma}\left(\sin^2\theta\right)^2\right]\Big|_{\theta=\pi/2}$$
$$=\left(r^2+a^2\right)-r^2\frac{2Mr}{\Sigma|_{\theta=\pi/2}}-\frac{2Mra^2}{\Sigma|_{\theta=\pi/2}}+\frac{2Mra^2}{\Sigma|_{\theta=\pi/2}}$$
$$=r^2+a^2-2Mr$$
$$=\Delta$$

Here, $\Sigma|_{\theta=\pi/2}=r^2+a^2\cos^2\theta\Big|_{\theta=\pi/2}=r^2$.
Method 2 (preferred):

$$\begin{aligned}g_{t\phi}^2-g_{\phi\phi}g_{tt}&=\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2-\left(-\frac{A}{\Sigma}\sin^2\theta\right)\left(1-\frac{2Mr}{\Sigma}\right)\\&=\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2-\left(-\frac{A}{\Sigma}\sin^2\theta\right)\left(\frac{\Sigma}{\Sigma}-\frac{2Mr}{\Sigma}\right)\\&=\frac{\sin^2\theta}{\Sigma^2}\left[\left(2Mar\right)^2\sin^2\theta+A\left(\Sigma-2Mr\right)\right]\\&=\Delta\sin^2\theta\end{aligned}$$

where

$$\begin{aligned}\left(2Mar\right)^2\sin^2\theta+A\left(\Sigma-2Mr\right)&=\left(2Mar\right)^2\sin^2\theta+A\left(\Sigma-2Mr\right)\\&=\left(2Mar\right)^2\sin^2\theta+\left[\left(r^2+a^2\right)\Sigma+2Mra^2\sin^2\theta\right]\left(\Sigma-2Mr\right)\\&=\left(2Mar\right)^2\sin^2\theta+\left(r^2+a^2\right)\Sigma\left(\Sigma-2Mr\right)+2Mra^2\sin^2\theta\left(\Sigma-2Mr\right)\\&=\left(r^2+a^2\right)\Sigma\left(\Sigma-2Mr\right)+\left(2Mar\right)^2\sin^2\theta+2Mra^2\sin^2\theta\left(\Sigma-2Mr\right)\\&=\left(r^2+a^2\right)\Sigma\left(\Sigma-2Mr\right)+2Mra^2\sin^2\theta\left[2Mr+\left(\Sigma-2Mr\right)\right]\\&=\Sigma\left[\left(r^2+a^2\right)\left(\Sigma-2Mr\right)+2Mra^2\sin^2\theta\right]\\&=\Sigma\left[\left(r^2+a^2\right)\left(\Delta-a^2\sin^2\theta\right)+2Mra^2\sin^2\theta\right]\\&=\Sigma\left[\left(r^2+a^2\right)\Delta-\left(r^2+a^2\right)a^2\sin^2\theta+2Mra^2\sin^2\theta\right]\\&=\Sigma\left[\left(r^2+a^2\right)\Delta-\left(r^2+a^2-2Mr\right)a^2\sin^2\theta\right]\\&=\Sigma\left[\left(r^2+a^2\right)\Delta-\Delta a^2\sin^2\theta\right]\\&=\Sigma\Delta\left[r^2+a^2\left(1-\sin^2\theta\right)\right]\\&=\Sigma\Delta\left(r^2+a^2\cos^2\theta\right)\\&=\Sigma^2\Delta\end{aligned}$$

where

$$\begin{aligned}\Sigma - 2Mr &= r^2 + a^2\cos^2\theta - 2Mr \\ &= r^2 + a^2\left(1-\sin^2\theta\right) - 2Mr \\ &= r^2 + a^2 - 2Mr - a^2\sin^2\theta \\ &= \Delta - a^2\sin^2\theta\end{aligned}$$

This gives

$$\begin{aligned}\left.\frac{\sqrt{g_{t\phi}^2 - g_{\phi\phi}g_{tt}}}{g_{\phi\phi}}\right|_{\theta=\pi/2} &= \left.\frac{\sqrt{\left(\frac{2Mar}{\Sigma}\sin^2\theta\right)^2 - \left(-\frac{A}{\Sigma}\sin^2\theta\right)\left(1-\frac{2Mr}{\Sigma}\right)}}{-\frac{A}{\Sigma}\sin^2\theta}\right|_{\theta=\pi/2} \\ &= \left.\frac{\sqrt{\Delta\sin^2\theta}}{-\frac{A}{\Sigma}\sin^2\theta}\right|_{\theta=\pi/2} \\ &= -\frac{\Sigma\sqrt{\Delta}}{A}\end{aligned}$$

We obtain

$$\begin{aligned}\Omega_{\pm}' &= -\frac{g_{t\phi}}{g_{\phi\phi}} \pm \frac{\sqrt{g_{t\phi}^2 - g_{\phi\phi}g_{tt}}}{g_{\phi\phi}} \\ &= \omega_d \mp \frac{\Sigma\sqrt{\Delta}}{A}\end{aligned}$$

Define

$$\Omega_{\pm} = \omega_d \pm \frac{\Sigma\sqrt{\Delta}}{A}$$

The condition $r > r_+$ is imposed because the orbit must lie outside the outer horizon of the black hole.

$$\begin{aligned}g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\big|_{\theta=\pi/2} &= g_{\phi\phi}\left(\frac{g_{tt}}{g_{\phi\phi}} + 2\Omega\frac{g_{t\phi}}{g_{\phi\phi}} + \Omega^2\right) \\ &= g_{\phi\phi}\left(\frac{g_{tt}}{g_{\phi\phi}} - 2\Omega\omega_d + \Omega^2\right) \\ &= g_{\phi\phi}\left[\frac{g_{tt}}{g_{\phi\phi}} + \left(\Omega-\omega_d\right)^2 - \omega_d^2\right] \\ &= \frac{g_{\phi\phi}g_{tt}}{g_{\phi\phi}} + g_{\phi\phi}\left(\Omega-\omega_d\right)^2 - g_{\phi\phi}\left(-\frac{g_{t\phi}}{g_{\phi\phi}}\right)^2 \\ &= g_{\phi\phi}\left(\Omega-\omega_d\right)^2 + \frac{g_{\phi\phi}g_{tt} - g_{t\phi}^2}{g_{\phi\phi}} \\ &= -\frac{A}{\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)^2 + \frac{-\Delta\sin^2\theta}{-\frac{A}{\Sigma}\sin^2\theta} \\ &= -\frac{A}{\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)^2 + \frac{\Delta\Sigma}{A} \\ &= \frac{\Delta\Sigma}{A}\left[1 - \frac{A^2}{\Delta\Sigma^2}\sin^2\theta\left(\Omega-\omega_d\right)^2\right]\Big|_{\theta=\pi/2} \\ &= \frac{\Delta\Sigma}{A}\left[1 - \frac{A^2}{\Delta\Sigma^2}\left(\Omega-\omega_d\right)^2\right]\end{aligned}$$

$$\begin{aligned}C\left(\Omega\right) &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right)^{-1/2} \\ &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\big|_{\theta=\pi/2}\right)^{-1/2} \\ &= \left\{\frac{\Delta\Sigma}{A}\left[1 - \frac{A^2}{\Delta\Sigma^2}\left(\Omega-\omega_d\right)^2\right]\right\}^{-1/2}\end{aligned}$$

Alternatively,

$$\begin{aligned} C(\Omega) &= \left\{ \frac{\Delta\Sigma|_{\theta=\pi/2}}{A} \left[1 - \frac{A^2 \sin^2\theta|_{\theta=\pi/2}}{\Delta\Sigma|^2_{\theta=\pi/2}} (\Omega - \omega_d)^2 \right] \right\}^{-1/2} \\ &= \left\{ \frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4} (\Omega - \omega_d)^2 \right] \right\}^{-1/2} \end{aligned}$$

For Eq. (16),

$$\begin{cases} t' = t \\ r' = r \\ \theta' = \theta \\ \phi' = \phi - \Omega t \end{cases}$$

$$\begin{aligned} w^{'\mu} &= \left(\frac{dt'}{d\tau}, \frac{dr'}{d\tau}, \frac{d\theta'}{d\tau}, \frac{d\phi'}{d\tau} \right) \\ &= \left(\frac{dt}{d\tau}, \frac{dr}{d\tau}, \frac{d\theta}{d\tau}, \frac{d(\phi - \Omega t)}{d\tau} \right) \\ &= \left(\frac{dt}{d\tau}, \frac{dr}{d\tau}, \frac{d\theta}{d\tau}, \frac{d\phi}{d\tau} - \Omega \frac{dt}{d\tau} \right) \\ &= \left(\frac{dt}{d\tau}, 0, 0, \Omega \frac{dt}{d\tau} - \Omega \frac{dt}{d\tau} \right) \\ &= C(\Omega)(1, 0, 0, 0) \\ &= C(\Omega)\, \delta^\mu_t \end{aligned}$$

Here, $\frac{d\phi}{d\tau} = \Omega \frac{dt}{d\tau}$ and $\frac{dt}{d\tau} = C(\Omega)$. A scalar, such as the inner product of the four-velocity $w^\mu w_\mu = 1$, is invariant under coordinate transformations. The transformed four-velocity $w^{'\mu}$ therefore does not need to be renormalized. Specifically, as shown above, the four-velocity becomes

$$w^{'\mu} = \frac{dt'}{d\tau} \delta^\mu_t$$

The normalization condition $w^{'\mu} w^{'}_{\mu} = 1$ gives

$$\frac{dt'}{d\tau} = \left(g^{'}_{tt} \right)^{-1/2}$$

We also have the result derived above:

$$g^{'}_{tt} = g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}$$

Hence,

$$\begin{aligned} \frac{dt'}{d\tau} &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi} \right)^{-1/2} \\ &= C(\Omega) \\ &= \frac{dt}{d\tau} \end{aligned}$$

The arc length is $\Delta l = r\Delta\theta$, so $dx = rd\phi'$ and $dz = rd\theta' = rd\theta$. We obtain

$$\begin{cases} d\phi' = \frac{1}{r} dx \\ d\theta' = \frac{1}{r} dz \\ dr' = dy \\ dt' = dt \end{cases}$$

$$
\begin{aligned}
ds^2 &= g^{'}_{\mu\nu} dx^{'\mu} dx^{'\nu} \\
&= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt'^2 + 2\left(g_{t\phi} + \Omega g_{\phi\phi}\right) dt' d\phi^{'} + g_{rr} dr'^2 + g_{\theta\theta} d\theta'^2 + g_{\phi\phi} d\phi^{'2} \\
&= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt^2 + 2\left(g_{t\phi} + \Omega g_{\phi\phi}\right) dt \left(\frac{1}{r} dx\right) + g_{rr} dy^2 + g_{\theta\theta} \left(\frac{1}{r} dz\right)^2 + g_{\phi\phi} \left(\frac{1}{r} dx\right)^2 \\
&= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt^2 + 2 g_{\phi\phi} \left(\frac{g_{t\phi}}{g_{\phi\phi}} + \Omega\right) dt \left(\frac{1}{r} dx\right) - \frac{\Sigma}{\Delta} dy^2 - \Sigma \left(\frac{1}{r} dz\right)^2 - \frac{A}{\Sigma} \sin^2\theta \left(\frac{1}{r} dx\right)^2 \\
&= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt^2 + 2\left(-\frac{A}{\Sigma} \sin^2\theta\right) \left(-\omega_d + \Omega\right) dt \left(\frac{1}{r} dx\right) - \frac{\Sigma}{\Delta} dy^2 - \Sigma \left(\frac{1}{r} dz\right)^2 - \frac{A}{\Sigma} \sin^2\theta \left(\frac{1}{r} dx\right)^2
\end{aligned}
$$

When $\theta = \pi/2$, we have $\Sigma = r^2$, and

$$
\begin{aligned}
ds^2 &= C^{-2}(\Omega)\, dt^2 - 2\frac{A}{r^3}(\Omega - \omega_d)\, dt dx - \frac{r^2}{\Delta} dy^2 - dz^2 - \frac{A}{r^4} dx^2 \\
&= \hat{g}_{tt} dt^2 + 2\hat{g}_{tx} dt dx + \hat{g}_{yy} dy^2 + \hat{g}_{zz} dz^2 + \hat{g}_{xx} dx^2
\end{aligned}
$$

$$
\begin{aligned}
\hat{g}_{\mu\nu} &= \begin{pmatrix} C^{-2}(\Omega) & -\frac{A}{r^3}(\Omega - \omega_d) & 0 & 0 \\ -\frac{A}{r^3}(\Omega - \omega_d) & -\frac{A}{r^4} & 0 & 0 \\ 0 & 0 & -\frac{r^2}{\Delta} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
&= \begin{pmatrix} \hat{g}_{tt} & \hat{g}_{tx} & 0 & 0 \\ \hat{g}_{xt} & \hat{g}_{xx} & 0 & 0 \\ 0 & 0 & \hat{g}_{yy} & 0 \\ 0 & 0 & 0 & \hat{g}_{zz} \end{pmatrix} \\
&= \begin{pmatrix} A & O \\ O & B \end{pmatrix}
\end{aligned}
$$

Here, $\hat{g}_{tx} = \hat{g}_{xt}$.

$$
\begin{aligned}
\hat{g}_{tt}\hat{g}_{xx} - \hat{g}_{tx}^2 &= C^{-2}(\Omega) \left(-\frac{A}{r^4}\right) - \left[\frac{A}{r^3}(\Omega - \omega_d)\right]^2 \\
&= \frac{\Delta\Sigma}{A} \left[1 - \frac{A^2}{\Delta\Sigma^2}(\Omega - \omega_d)^2\right] \left(-\frac{A}{r^4}\right) - \left[\frac{A}{r^3}(\Omega - \omega_d)\right]^2 \\
&= \frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4}(\Omega - \omega_d)^2\right] \left(-\frac{A}{r^4}\right) - \left[\frac{A}{r^3}(\Omega - \omega_d)\right]^2 \\
&= -\frac{\Delta}{r^2} + \frac{A^2}{r^6}(\Omega - \omega_d)^2 - \frac{A^2}{r^6}(\Omega - \omega_d)^2 \\
&= -\frac{\Delta}{r^2}
\end{aligned}
$$

Here, for an equatorial orbit, $\theta = \pi/2$ and $\Sigma = r^2$.

$$
\begin{aligned}
\det(\hat{g}_{\mu\nu}) &= \begin{vmatrix} \hat{g}_{tt} & \hat{g}_{tx} & 0 & 0 \\ \hat{g}_{xt} & \hat{g}_{xx} & 0 & 0 \\ 0 & 0 & \hat{g}_{yy} & 0 \\ 0 & 0 & 0 & \hat{g}_{zz} \end{vmatrix} \\
&= \begin{vmatrix} A & O \\ O & B \end{vmatrix} \\
&= \det(A) \cdot \det(B) \\
&= \left(\hat{g}_{tt}\hat{g}_{xx} - \hat{g}_{tx}^2\right) \times \hat{g}_{yy} \times \hat{g}_{zz} \\
&= -\frac{\Delta}{r^2} \times \left(-\frac{r^2}{\Delta}\right) \times (-1) \\
&= -1
\end{aligned}
$$

This is equal to the determinant of the Minkowski metric, $\det(\eta_{\mu\nu}) = -1$, for an equatorial orbit ($\theta = \pi/2$).

Inverting the metric:

$$\hat{g}^{\mu\alpha}\hat{g}_{\alpha\nu} = \delta^{\mu}_{\nu}$$

Equivalently, in matrix notation,

$$\hat{g}^{\mu\nu} = (\hat{g}_{\mu\nu})^{-1}$$

$$(\hat{g}_{\mu\nu})^{-1} = \begin{pmatrix} A^{-1} & O \\ O & B^{-1} \end{pmatrix}$$

$$\begin{pmatrix} \hat{g}_{tt} & \hat{g}_{tx} \\ \hat{g}_{xt} & \hat{g}_{xx} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$
$$\begin{pmatrix} a\hat{g}_{tt} + c\hat{g}_{tx} & b\hat{g}_{tt} + d\hat{g}_{tx} \\ a\hat{g}_{xt} + c\hat{g}_{xx} & b\hat{g}_{xt} + d\hat{g}_{xx} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{cases} a\hat{g}_{tt} + c\hat{g}_{tx} = 1 \\ b\hat{g}_{tt} + d\hat{g}_{tx} = 0 \\ a\hat{g}_{xt} + c\hat{g}_{xx} = 0 \\ b\hat{g}_{xt} + d\hat{g}_{xx} = 1 \end{cases}$$

$$\begin{aligned} (a\hat{g}_{tt} + c\hat{g}_{tx}) + (a\hat{g}_{xt} + c\hat{g}_{xx}) &= 1 \\ a\left(\hat{g}_{tt} + \hat{g}_{xt}\right) + \frac{1 - a\hat{g}_{tt}}{\hat{g}_{tx}}\left(\hat{g}_{tx} + \hat{g}_{xx}\right) &= 1 \\ a\left(\hat{g}_{tt} + \hat{g}_{xt}\right) + \frac{1}{\hat{g}_{tx}}\left(\hat{g}_{tx} + \hat{g}_{xx}\right) - \frac{a\hat{g}_{tt}}{\hat{g}_{tx}}\left(\hat{g}_{tx} + \hat{g}_{xx}\right) &= 1 \\ a\left(\hat{g}_{tt} + \hat{g}_{xt}\right) + \left(1 + \frac{\hat{g}_{xx}}{\hat{g}_{tx}}\right) - a\hat{g}_{tt}\left(1 + \frac{\hat{g}_{xx}}{\hat{g}_{tx}}\right) &= 1 \\ a\left[\left(\hat{g}_{tt} + \hat{g}_{xt}\right) - \hat{g}_{tt}\left(1 + \frac{\hat{g}_{xx}}{\hat{g}_{tx}}\right)\right] &= 1 - \left(1 + \frac{\hat{g}_{xx}}{\hat{g}_{tx}}\right) \\ a &= \frac{1 - \left(1 + \frac{\hat{g}_{xx}}{\hat{g}_{tx}}\right)}{\left(\hat{g}_{tt} + \hat{g}_{xt}\right) - \hat{g}_{tt}\left(1 + \frac{\hat{g}_{xx}}{\hat{g}_{tx}}\right)} \\ a &= \frac{-\hat{g}_{xx}}{\hat{g}_{tx}\hat{g}_{xt} - \hat{g}_{tt}\hat{g}_{xx}} \end{aligned}$$

$$\begin{aligned} c &= \frac{-a\hat{g}_{xt}}{\hat{g}_{xx}} \\ &= -\left(\frac{-\hat{g}_{xx}}{\hat{g}_{tx}\hat{g}_{xt} - \hat{g}_{tt}\hat{g}_{xx}}\right)\frac{\hat{g}_{xt}}{\hat{g}_{xx}} \\ &= \frac{\hat{g}_{xt}}{\hat{g}_{tx}\hat{g}_{xt} - \hat{g}_{tt}\hat{g}_{xx}} \end{aligned}$$

$$\begin{aligned} (b\hat{g}_{tt} + d\hat{g}_{tx}) + (b\hat{g}_{xt} + d\hat{g}_{xx}) &= 1 \\ b\left(\hat{g}_{tt} + \hat{g}_{xt}\right) + \frac{-b\hat{g}_{tt}}{\hat{g}_{tx}}\left(\hat{g}_{tx} + \hat{g}_{xx}\right) &= 1 \\ b\left(\hat{g}_{tt} + \hat{g}_{xt}\right) - b\left(\hat{g}_{tt} + \frac{\hat{g}_{tt}\hat{g}_{xx}}{\hat{g}_{tx}}\right) &= 1 \\ b\left(\hat{g}_{xt} - \frac{\hat{g}_{tt}\hat{g}_{xx}}{\hat{g}_{tx}}\right) &= 1 \\ b &= \frac{\hat{g}_{tx}}{\hat{g}_{tx}\hat{g}_{xt} - \hat{g}_{tt}\hat{g}_{xx}} \end{aligned}$$

$$\begin{aligned} d &= \frac{-b\hat{g}_{tt}}{\hat{g}_{tx}} \\ &= \frac{-\hat{g}_{tx}}{\hat{g}_{tx}\hat{g}_{xt} - \hat{g}_{tt}\hat{g}_{xx}}\frac{\hat{g}_{tt}}{\hat{g}_{tx}} \\ &= \frac{-\hat{g}_{tt}}{\hat{g}_{tx}\hat{g}_{xt} - \hat{g}_{tt}\hat{g}_{xx}} \end{aligned}$$

$$
\begin{aligned}
A^{-1} &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\
&= \frac{1}{\hat{g}_{tx}\hat{g}_{xt} - \hat{g}_{tt}\hat{g}_{xx}} \begin{pmatrix} -\hat{g}_{xx} & \hat{g}_{tx} \\ \hat{g}_{xt} & -\hat{g}_{tt} \end{pmatrix} \\
&= \frac{1}{\hat{g}_{tx}^2 - \hat{g}_{tt}\hat{g}_{xx}} \begin{pmatrix} -\hat{g}_{xx} & \hat{g}_{tx} \\ \hat{g}_{tx} & -\hat{g}_{tt} \end{pmatrix}
\end{aligned}
$$

$$
\begin{pmatrix} \hat{g}_{yy} & 0 \\ 0 & \hat{g}_{zz} \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
$$

$$
\begin{pmatrix} \hat{g}_{yy}\alpha & \hat{g}_{yy}\beta \\ \hat{g}_{zz}\gamma & \hat{g}_{zz}\delta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
$$

$$
\begin{cases} \alpha = \frac{1}{\hat{g}_{yy}} \\ \beta = 0 \\ \gamma = 0 \\ \delta = \frac{1}{\hat{g}_{zz}} \end{cases}
$$

$$
\begin{aligned}
B^{-1} &= \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{\hat{g}_{yy}} & 0 \\ 0 & \frac{1}{\hat{g}_{zz}} \end{pmatrix}
\end{aligned}
$$

We obtain the inverse metric matrix:

$$
\begin{aligned}
(\hat{g}_{\mu\nu})^{-1} &= \begin{pmatrix} A^{-1} & O \\ O & B^{-1} \end{pmatrix} \\
&= \begin{pmatrix} \frac{-\hat{g}_{xx}}{\hat{g}_{tx}^2 - \hat{g}_{tt}\hat{g}_{xx}} & \frac{\hat{g}_{tx}}{\hat{g}_{tx}^2 - \hat{g}_{tt}\hat{g}_{xx}} & 0 & 0 \\ \frac{\hat{g}_{tx}}{\hat{g}_{tx}^2 - \hat{g}_{tt}\hat{g}_{xx}} & \frac{-\hat{g}_{tt}}{\hat{g}_{tx}^2 - \hat{g}_{tt}\hat{g}_{xx}} & 0 & 0 \\ 0 & 0 & \frac{1}{\hat{g}_{yy}} & 0 \\ 0 & 0 & 0 & \frac{1}{\hat{g}_{zz}} \end{pmatrix} \\
&= \begin{pmatrix} \frac{\frac{A}{r^4}}{\frac{\Delta}{r^2}} & \frac{-\frac{A}{r^3}(\Omega - \omega_d)}{\frac{\Delta}{r^2}} & 0 & 0 \\ \frac{-\frac{A}{r^3}(\Omega - \omega_d)}{\frac{\Delta}{r^2}} & \frac{-C^{-2}(\Omega)}{\frac{\Delta}{r^2}} & 0 & 0 \\ 0 & 0 & \frac{1}{-\frac{r^2}{\Delta}} & 0 \\ 0 & 0 & 0 & \frac{1}{-1} \end{pmatrix} \\
&= \begin{pmatrix} \frac{A}{\Delta r^2} & \frac{A}{\Delta r}(\omega_d - \Omega) & 0 & 0 \\ \frac{A}{\Delta r}(\omega_d - \Omega) & -\frac{r^2}{\Delta}C^{-2}(\Omega) & 0 & 0 \\ 0 & 0 & -\frac{\Delta}{r^2} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}
\end{aligned}
$$

When the metric components are approximated as constants, the d'Alembertian is

$$
\begin{aligned}
\partial_s^2 &= \hat{g}^{\mu\nu}\partial_\mu\partial_\nu \\
&= \frac{A}{\Delta r^2}\partial_t^2 + 2\frac{A}{\Delta r}(\omega_d - \Omega)\partial_t\partial_x - \frac{r^2}{\Delta}C^{-2}(\Omega)\partial_x^2 - \frac{\Delta}{r^2}\partial_y^2 - \partial_z^2
\end{aligned}
$$

where $\partial_\mu\partial_\nu = \left(\frac{\partial}{\partial x^\mu}\right)\left(\frac{\partial}{\partial x^\nu}\right)$.

The components of the four-velocity transform contravariantly under a coordinate transformation. This follows because the coordinate differentials dx^μ transform contravariantly, whereas the proper-time interval $d\tau$ is a scalar. Under the transformation from the old coordinates x'^μ to the new coordinates $\hat{x}^\alpha$, the chain rule gives

$$
d\hat{x}^\alpha = \frac{\partial \hat{x}^\alpha}{\partial x'^\mu} dx'^\mu
$$

This is the transformation law relating coordinate differentials in the two systems. Dividing both sides by $d\tau$ gives

$$
\frac{d\hat{x}^\alpha}{d\tau} = \frac{\partial \hat{x}^\alpha}{\partial x'^\mu}\frac{dx'^\mu}{d\tau}
$$

or equivalently,

$$\hat{w}^{\alpha} = \frac{\partial \hat{x}^{\alpha}}{\partial x^{'\mu}} w^{'\mu}$$

From

$$\begin{cases} d\phi' = \frac{1}{r} dx \\ d\theta' = \frac{1}{r} dz \\ dr' = dy \\ dt' = dt \end{cases}$$

we obtain

$$\begin{cases} \frac{\partial x}{\partial \phi'} = r \\ \frac{\partial z}{\partial \theta'} = r \\ \frac{\partial y}{\partial r'} = 1 \\ \frac{\partial t}{\partial t'} = 1 \end{cases}$$

$$\begin{aligned} \hat{w}^{\hat{t}} &= \frac{\partial \hat{t}}{\partial x^{'\mu}} w^{'\mu} \\ &= \frac{\partial \hat{t}}{\partial t'} w^{'t'} + \frac{\partial \hat{t}}{\partial r'} w^{'r'} + \frac{\partial \hat{t}}{\partial \theta'} w^{'\theta'} + \frac{\partial \hat{t}}{\partial \phi'} w^{'\phi'} \\ &= \frac{\partial t}{\partial t} C(\Omega) + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 \\ &= C(\Omega) \end{aligned}$$

Here, $\hat{t} = t^{'} = t$. Since $\hat{t} = t^{'}$ does not depend on $r^{'}, \theta^{'}$ or $\phi^{'}$, we have $\partial \hat{t}/\partial r^{'} = \partial \hat{t}/\partial \theta^{'} = \partial \hat{t}/\partial \phi^{'} = 0$.

$$\begin{aligned} \hat{w}^{x} &= \frac{\partial x}{\partial x^{'\mu}} w^{'\mu} \\ &= \frac{\partial x}{\partial t'} w^{'t'} + \frac{\partial x}{\partial r'} w^{'r'} + \frac{\partial x}{\partial \theta'} w^{'\theta'} + \frac{\partial x}{\partial \phi'} w^{'\phi'} \\ &= 0 \cdot w^{'t'} + \frac{\partial x}{\partial r'} \cdot 0 + \frac{\partial x}{\partial \theta'} \cdot 0 + r \cdot 0 \\ &= 0 \end{aligned}$$

Similarly,

$$\hat{w}^{y} = \hat{w}^{z} = 0$$

Thus,

$$\hat{w}^{\mu} = w^{'\mu}$$

II. The Constant-Metric Approximation

Taylor expansion of a multivariable function:

$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0) \cdot (\boldsymbol{x} - \boldsymbol{x}_0) + \mathcal{O}\left(\|\boldsymbol{x} - \boldsymbol{x}_0\|^2\right)$$

For example, let $\boldsymbol{x} = (a, b, c)$ and $\boldsymbol{x}_0 = (a_0, b_0, c_0)$. Then

$$\nabla f(\boldsymbol{x}_0) = \left(\frac{\partial f(\boldsymbol{x}_0)}{\partial x^1}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x^2}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x^3} \right)$$

$$[\nabla f(\boldsymbol{x}_0)]_i = \partial_i f(\boldsymbol{x}_0) = \left. \frac{\partial f(a, b, c)}{\partial x^i} \right|_{\boldsymbol{x} = \boldsymbol{x}_0}$$

and

$$\|\boldsymbol{x} - \boldsymbol{x}_0\| = \sqrt{(a - a_0)^2 + (b - b_0)^2 + (c - c_0)^2}$$

Define

$$\delta \boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}_0 = (a - a_0, b - b_0, c - c_0)$$

Then

$$\nabla f(\boldsymbol{x}_0)\cdot(\boldsymbol{x}-\boldsymbol{x}_0)=\left.\frac{\partial f(a,b,c)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}\cdot\delta x^i$$
$$=\left.\frac{\partial f(a,b,c)}{\partial a}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}(a-a_0)+\left.\frac{\partial f(a,b,c)}{\partial b}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}(b-b_0)+\left.\frac{\partial f(a,b,c)}{\partial c}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}(c-c_0)$$

If terms through second order are retained, the expansion is

$$f(\boldsymbol{x})=f(\boldsymbol{x}_0)+\nabla f(\boldsymbol{x}_0)\cdot(\boldsymbol{x}-\boldsymbol{x}_0)+\frac{1}{2}(\boldsymbol{x}-\boldsymbol{x}_0)^T H(\boldsymbol{x}_0)(\boldsymbol{x}-\boldsymbol{x}_0)+\mathcal{O}\left(\|\boldsymbol{x}-\boldsymbol{x}_0\|^3\right)$$

where H is the Hessian matrix, whose entries are

$$H_{ij}=\frac{\partial^2 f}{\partial x_i\partial x_j}$$

Equivalently,

$$f\Big(a,b,c\Big)=f\Big(a,b,c\Big)\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}+\sum_{i=1}^{3}\left.\frac{\partial f\Big(a,b,c\Big)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}\Big(x^i-x_0^i\Big)+\frac{1}{2}\sum_{i,j=1}^{3}\left.\frac{\partial^2 f\Big(a,b,c\Big)}{\partial x^i\partial x^j}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}$$
$$\Big(x^i-x_0^i\Big)\Big(x^j-x_0^j\Big)+\mathcal{O}\Bigg(\Bigg(\sqrt{\sum_{i=1}^{3}\Big(x^i-x_0^i\Big)^2}\Bigg)^3\Bigg)$$

By Taylor's theorem with the mean-value form of the remainder, $f(a,b,c)$ can also be written as

$$f(a,b,c)=f(a,b,c)\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}+\sum_{i=1}^{3}\left.\frac{\partial f(a,b,c)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}\left(x^i-x_0^i\right)+\mathcal{R}$$

where

$$\mathcal{R}=\frac{1}{2}\sum_{i,j=1}^{3}\left.\frac{\partial^2 f(a,b,c)}{\partial x^i\partial x^j}\right|_{\boldsymbol{x}=\boldsymbol{\xi}}\left(x^i-x_0^i\right)\left(x^j-x_0^j\right)$$

Here, $\boldsymbol{\xi}=(\xi_1,\xi_2,\xi_3)$, and $\xi_1,\ \xi_2,\ \xi_3$ lie between a_0 and a, b_0 and b, and c_0 and c, respectively (or, equivalently, $\boldsymbol{\xi}=\boldsymbol{x}_0+s(\boldsymbol{x}-\boldsymbol{x}_0),\ 0<s<1$).

The Taylor expansion of the metric components $\hat{g}_{\mu\nu}$ about $\boldsymbol{x}_0$ is

$$\hat{g}_{\mu\nu}(t,x,y,z)=\hat{g}_{\mu\nu}(t,x,y,z)\big|_{\boldsymbol{x}=\boldsymbol{x}_0}+\sum_{i=1}^{3}\left.\frac{\partial\hat{g}_{\mu\nu}(t,x,y,z)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}\left(x^i-x_0^i\right)+\mathcal{O}\left(\|\boldsymbol{x}-\boldsymbol{x}_0\|^2\right)$$
$$=\hat{g}_{\mu\nu}(t,x_0,y_0,z_0)+\sum_{i=1}^{3}\left.\frac{\partial\hat{g}_{\mu\nu}(t,x,y,z)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}\cdot\delta x^i+\mathcal{O}\left(\|\delta\boldsymbol{x}\|^2\right)$$

or

$$\hat{g}_{\mu\nu}\Big(t,x,y,z\Big)=\hat{g}_{\mu\nu}\Big(t,x,y,z\Big)\big|_{\boldsymbol{x}=\boldsymbol{x}_0}+\sum_{i=1}^{3}\left.\frac{\partial\hat{g}_{\mu\nu}\Big(t,x,y,z\Big)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0}\Big(x^i-x_0^i\Big)+\frac{1}{2}\sum_{i,j=1}^{3}$$
$$\left.\frac{\partial^2\hat{g}_{\mu\nu}\Big(t,x,y,z\Big)}{\partial x^i\partial x^j}\right|_{\boldsymbol{x}=\boldsymbol{\xi}}\Big(x^i-x_0^i\Big)\Big(x^j-x_0^j\Big)$$

where $\boldsymbol{\xi}=(\xi_1,\xi_2,\xi_3)$, and $\xi_1,\ \xi_2,\ \xi_3$ lie between x_0 and x, y_0 and y, and z_0 and z, respectively (or, equivalently, $\boldsymbol{\xi}=\boldsymbol{x}_0+s(\boldsymbol{x}-\boldsymbol{x}_0),\ 0<s<1$). Here, $\boldsymbol{x}=(x,y,z)$ and $\boldsymbol{x}_0=(x_0,y_0,z_0)$. Define $\delta x^i=x^i-x_0^i$, or equivalently $x^i=x_0^i+\delta x^i$; explicitly, $\delta x=x-x_0,\delta y=y-y_0,\delta z=z-z_0$. Also define $\delta\boldsymbol{x}=\boldsymbol{x}-\boldsymbol{x}_0$. Let $\boldsymbol{x}_0=(x_0,y_0,z_0)$ correspond to $\boldsymbol{x}_0=(r_0,\theta_0,\phi_0)$ in the coordinate system (t,r,θ,ϕ). Under the coordinate transformations

$$\begin{cases}t'=t\\ r'=r\\ \theta'=\theta\\ \phi'=\phi-\Omega t\end{cases}$$

and

$$\begin{cases} d\bar{t} = dt' = dt \\ dx = r_0 d\phi' = r_0 d\left(\phi - \Omega t\right) = r_0 d\phi - \Omega r_0 dt \\ dy = dr' = dr \\ dz = r_0 d\theta' = r_0 d\theta \end{cases}$$

the variations near the point $\boldsymbol{x}_0 = (r_0, \theta_0, \phi_0)$ are

$$\begin{cases} \delta x = \left(r_0\phi - \Omega r_0 t\right) - \left(r_0\phi_0 - \Omega r_0 t\right) = r_0\left(\phi - \phi_0\right) \\ \delta y = r - r_0 \\ \delta z = r_0\left(\theta - \theta_0\right) \end{cases}$$

Here, the angular velocity Ω of the Casimir cavity is treated as a constant. Applying the coordinate transformation

$$\begin{cases} d\phi' = \frac{1}{r_0} dx \\ d\theta' = \frac{1}{r_0} dz \\ dr' = dy \\ dt' = dt \end{cases}$$

based at the point $\boldsymbol{x}_0 = (r_0, \theta_0, \phi_0)$ gives

$$\begin{aligned} ds^2 &= g'_{\mu\nu} dx'^{\mu} dx'^{\nu} \\ &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt'^2 + 2\left(g_{t\phi} + \Omega g_{\phi\phi}\right) dt' d\phi' + g_{rr} dr'^2 + g_{\theta\theta} d\theta'^2 + g_{\phi\phi} d\phi'^2 \\ &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt^2 + 2\left(g_{t\phi} + \Omega g_{\phi\phi}\right) dt \left(\frac{1}{r_0} dx\right) + g_{rr} dy^2 + g_{\theta\theta} \left(\frac{1}{r_0} dz\right)^2 + g_{\phi\phi} \left(\frac{1}{r_0} dx\right)^2 \\ &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt^2 + 2\left(-\frac{A}{\Sigma}\sin^2\theta\right)\left(-\omega_d + \Omega\right) dt \left(\frac{1}{r_0} dx\right) - \frac{\Sigma}{\Delta} dy^2 - \Sigma \left(\frac{1}{r_0} dz\right)^2 - \frac{A}{\Sigma}\sin^2\theta \left(\frac{1}{r_0} dx\right)^2 \\ &= \left(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi}\right) dt^2 - 2\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega - \omega_d\right) dt dx - \frac{\Sigma}{\Delta} dy^2 - \frac{\Sigma}{r_0^2} dz^2 - \frac{A}{r_0^2\Sigma}\sin^2\theta dx^2 \\ &= C^{-2}\left(\Omega; r, \theta\right) dt^2 - 2\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega - \omega_d\right) dt dx - \frac{A}{r_0^2\Sigma}\sin^2\theta dx^2 - \frac{\Sigma}{\Delta} dy^2 - \frac{\Sigma}{r_0^2} dz^2 \\ &= \hat{g}_{tt} dt^2 + 2\hat{g}_{tx} dt dx + \hat{g}_{xx} dx^2 + \hat{g}_{yy} dy^2 + \hat{g}_{zz} dz^2 \end{aligned}$$

where

$$C\left(\Omega; r, \theta\right) = \left\{ \frac{\Delta\Sigma}{A}\left[1 - \frac{A^2\sin^2\theta}{\Delta\Sigma^2}\left(\Omega - \omega_d\right)^2\right]\right\}^{-1/2}$$

The coordinate r is a function of x, y, z.

$$\begin{aligned} \hat{g}_{\mu\nu} &= \begin{pmatrix} C^{-2}\left(\Omega; r, \theta\right) & -\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega - \omega_d\right) & 0 & 0 \\ -\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega - \omega_d\right) & -\frac{A}{r_0^2\Sigma}\sin^2\theta & 0 & 0 \\ 0 & 0 & -\frac{\Sigma}{\Delta} & 0 \\ 0 & 0 & 0 & -\frac{\Sigma}{r_0^2} \end{pmatrix} \\ &= \begin{pmatrix} \hat{g}_{tt} & \hat{g}_{tx} & 0 & 0 \\ \hat{g}_{xt} & \hat{g}_{xx} & 0 & 0 \\ 0 & 0 & \hat{g}_{yy} & 0 \\ 0 & 0 & 0 & \hat{g}_{zz} \end{pmatrix} \end{aligned}$$

Here, $\hat{g}_{tx} = \hat{g}_{xt}$.

$$\sum_{i=1}^{3} \left.\frac{\partial \hat{g}_{\mu\nu}\left(t, x, y, z\right)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0} \cdot \delta x^i$$

$$
= \begin{pmatrix}
\frac{\partial \hat{g}_{tt}}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial \hat{g}_{tt}}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial \hat{g}_{tt}}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & \frac{\partial \hat{g}_{tx}}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial \hat{g}_{tx}}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial \hat{g}_{tx}}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & 0 & 0 \\
\frac{\partial \hat{g}_{xt}}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial \hat{g}_{xt}}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial \hat{g}_{xt}}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & \frac{\partial \hat{g}_{xx}}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial \hat{g}_{xx}}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial \hat{g}_{xx}}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & 0 & 0 \\
0 & 0 & \frac{\partial \hat{g}_{yy}}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial \hat{g}_{yy}}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial \hat{g}_{yy}}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & 0 \\
0 & 0 & 0 & \frac{\partial \hat{g}_{zz}}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial \hat{g}_{zz}}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial \hat{g}_{zz}}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z
\end{pmatrix}
$$

$$
= \begin{pmatrix}
\frac{\partial C^{-2}\left(\Omega;r,\theta\right)}{\partial \phi'}\frac{\partial \phi'}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial C^{-2}\left(\Omega;r,\theta\right)}{\partial r'}\frac{\partial r'}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial C^{-2}\left(\Omega;r,\theta\right)}{\partial \theta'}\frac{\partial \theta'}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & \frac{\partial\left[-\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)\right]}{\partial \phi'}\frac{\partial \phi'}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial\left[-\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)\right]}{\partial r'}\frac{\partial r'}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial\left[-\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)\right]}{\partial \theta'}\frac{\partial \theta'}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & 0 & 0 \\
\frac{\partial\left[-\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)\right]}{\partial \phi'}\frac{\partial \phi'}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial\left[-\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)\right]}{\partial r'}\frac{\partial r'}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial\left[-\frac{A}{r_0\Sigma}\sin^2\theta\left(\Omega-\omega_d\right)\right]}{\partial \theta'}\frac{\partial \theta'}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & \frac{\partial\left[-\frac{A}{r_0^2\Sigma}\sin^2\theta\right]}{\partial \phi'}\frac{\partial \phi'}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial\left[-\frac{A}{r_0^2\Sigma}\sin^2\theta\right]}{\partial r'}\frac{\partial r'}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial\left[-\frac{A}{r_0^2\Sigma}\sin^2\theta\right]}{\partial \theta'}\frac{\partial \theta'}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & 0 & 0 \\
0 & 0 & \frac{\partial\left[-\frac{\Sigma}{\Delta}\right]}{\partial \phi'}\frac{\partial \phi'}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial\left[-\frac{\Sigma}{\Delta}\right]}{\partial r'}\frac{\partial r}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial\left[-\frac{\Sigma}{\Delta}\right]}{\partial \theta'}\frac{\partial \theta'}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z & 0 \\
0 & 0 & 0 & \frac{\partial\left[-\frac{\Sigma}{r_0^2}\right]}{\partial \phi'}\frac{\partial \phi'}{\partial x}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta x + \frac{\partial\left[-\frac{\Sigma}{r_0^2}\right]}{\partial r'}\frac{\partial r'}{\partial y}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta y + \frac{\partial\left[-\frac{\Sigma}{r_0^2}\right]}{\partial \theta'}\frac{\partial \theta'}{\partial z}\Big|_{\boldsymbol{x}=\boldsymbol{x}_0}\delta z
\end{pmatrix}
$$

Here,

$$\begin{aligned}\frac{\partial f\left(t,r,\theta,\phi\right)}{\partial\phi^{'}} &= \frac{\partial f\left(t,r,\theta,\phi^{'}+\Omega t\right)}{\partial\phi^{'}}\\ &= \frac{\partial f\left(t,r,\theta,\phi^{'}\left(\phi,t\right)+\Omega t\right)}{\partial\phi^{'}}\\ &= \frac{\partial f}{\partial\phi}\frac{\partial\phi}{\partial\phi^{'}}+\frac{\partial f}{\partial t}\frac{\partial t}{\partial\phi^{'}}\text{ (chain rule)}\\ &= \frac{\partial f}{\partial\phi}\frac{\partial\left(\phi^{'}+\Omega t\right)}{\partial\phi^{'}}+\frac{\partial f}{\partial t}\frac{\partial t}{\partial\phi^{'}}\\ &= \frac{\partial f}{\partial\phi}\left(\text{t is an independent variable, }\frac{\partial t}{\partial\phi^{'}}=0\right)\end{aligned}$$

(An additional, redundant intermediate step would give

$$\begin{aligned}\frac{\partial f\left(t,r,\theta,\phi\right)}{\partial\left(\phi-\Omega t\right)}\frac{\partial\left(\phi-\Omega t\right)}{\partial\phi^{'}}\frac{\partial\phi^{'}}{\partial x} &= \frac{\partial f\left(t,r,\theta,\psi+\Omega t\right)}{\partial\psi}\frac{\partial\psi}{\partial\left(\phi-\Omega t\right)}\frac{\partial\left(\phi-\Omega t\right)}{\partial\phi^{'}}\frac{\partial\phi^{'}}{\partial x}\\ &= \frac{\partial f\left(t,r,\theta,\psi\left(\phi,t\right)+\Omega t\right)}{\partial\psi}\frac{\partial\left(\phi-\Omega t\right)}{\partial\left(\phi-\Omega t\right)}\frac{\partial\left(\phi-\Omega t\right)}{\partial\left(\phi-\Omega t\right)}\frac{\partial\phi^{'}}{\partial x}\\ &= \left(\frac{\partial f}{\partial\phi}\frac{\partial\phi}{\partial\psi}+\frac{\partial f}{\partial t}\frac{\partial t}{\partial\psi}\right)\frac{\partial\phi^{'}}{\partial x}\\ &= \left(\frac{\partial f}{\partial\phi}\frac{\partial\left(\psi+\Omega t\right)}{\partial\psi}+\frac{\partial f}{\partial t}\frac{\partial t}{\partial\psi}\right)\frac{\partial\phi^{'}}{\partial x}\\ &= \frac{\partial f}{\partial\phi}\frac{\partial\phi^{'}}{\partial x}\end{aligned}$$

)

We obtain

$$\begin{aligned}\frac{\partial C^{-2}\left(\Omega;r,\theta\right)}{\partial\phi^{'}}\frac{\partial\phi^{'}}{\partial x} &= \frac{\partial\left\{\frac{\Delta\Sigma}{A}\left[1-\frac{A^{2}\sin^{2}\theta}{\Delta\Sigma^{2}}\left(\Omega-\omega_{d}\right)^{2}\right]\right\}}{\partial\phi^{'}}\cdot\frac{1}{r_{0}}\\ &= \frac{\partial\left\{\frac{\Delta\Sigma}{A}\left[1-\frac{A^{2}\sin^{2}\theta}{\Delta\Sigma^{2}}\left(\Omega-\omega_{d}\right)^{2}\right]\right\}}{\partial\phi}\cdot\frac{1}{r_{0}}=0\cdot\frac{1}{r_{0}}=0\end{aligned}$$

In other words, since none of the metric components $\hat{g}_{\mu\nu}$ depends on ϕ, for all μ,ν we have

$$\frac{\partial\hat{g}_{\mu\nu}}{\partial x}=\frac{\partial\hat{g}_{\mu\nu}}{\partial\phi^{'}}\frac{\partial\phi^{'}}{\partial x}=\frac{\partial\hat{g}_{\mu\nu}}{\partial\phi}\frac{\partial\phi^{'}}{\partial x}=0$$

Introduce the notation

$$\mathcal{D}f\equiv\frac{\partial f}{\partial r^{'}}\frac{\partial r^{'}}{\partial y}\delta y+\frac{\partial f}{\partial\theta^{'}}\frac{\partial\theta^{'}}{\partial z}\delta z=\frac{\partial f}{\partial r}\delta y+\frac{1}{r_{0}}\frac{\partial f}{\partial\theta}\delta z$$

$$Q\equiv\frac{A\sin^{2}\theta}{\Sigma}$$

The following partial derivatives were obtained using a Python program. The subscript 0 denotes evaluation at $\boldsymbol{x}=\boldsymbol{x}_{0}$. Write $s_{0}=\sin\theta_{0}$, $c_{0}=\cos\theta_{0}$, $\Sigma_{0}=r_{0}^{2}+a^{2}c_{0}^{2}$, $\Delta_{0}=r_{0}^{2}-2Mr_{0}+a^{2}$, and $U_{0}=1-a\Omega s_{0}^{2}$.

$$\partial_{x}\hat{g}|_{0}=0$$

$$R_{0}=\partial_{y}\hat{g}|_{0}=\begin{pmatrix}-\frac{2MU_{0}^{2}}{\Sigma_{0}}+\frac{4MU_{0}^{2}r_{0}^{2}}{\Sigma_{0}^{2}}-2\Omega^{2}r_{0}s_{0}^{2} & \frac{2s_{0}^{2}\left(\frac{MU_{0}a}{\Sigma_{0}}-\frac{2MU_{0}ar_{0}^{2}}{\Sigma_{0}^{2}}-\Omega r_{0}\right)}{r_{0}} & 0 & 0\\ \frac{2s_{0}^{2}\left(\frac{MU_{0}a}{\Sigma_{0}}-\frac{2MU_{0}ar_{0}^{2}}{\Sigma_{0}^{2}}-\Omega r_{0}\right)}{r_{0}} & -\frac{2s_{0}^{2}\left(\frac{Ma^{2}s_{0}^{2}}{\Sigma_{0}}-\frac{2Ma^{2}r_{0}^{2}s_{0}^{2}}{\Sigma_{0}^{2}}+r_{0}\right)}{r_{0}^{2}} & 0 & 0\\ 0 & 0 & -\frac{2\left(r_{0}+\frac{\Sigma_{0}(M-r_{0})}{\Delta_{0}}\right)}{\Delta_{0}} & 0\\ 0 & 0 & 0 & -\frac{2}{r_{0}}\end{pmatrix}$$

$$P_0 = \partial_z \hat{g}\Big|_0 =$$

$$\begin{pmatrix}
\begin{matrix} 2c_0 s_0 \Bigg(\frac{4M\Omega U_0 a r_0}{\Sigma_0} \\ -\frac{2MU_0^2 a^2 r_0}{\Sigma_0^2} - \Omega^2 \Big(a^2 + r_0^2 \Big) \Bigg) \end{matrix} \frac{}{r_0} & \begin{matrix} 2c_0 s_0 \Bigg(\frac{2MU_0 a r_0}{\Sigma_0} \\ + \frac{2Ma^2 r_0 s_0^2 \Big(-\Omega + \frac{U_0 a}{\Sigma_0} \Big)}{\Sigma_0} \\ -\Omega \Big(a^2 + r_0^2 \Big) \Bigg) \end{matrix} \frac{}{r_0^2} & 0 & 0 \\
\begin{matrix} 2c_0 s_0 \Bigg(\frac{2MU_0 a r_0}{\Sigma_0} \\ + \frac{2Ma^2 r_0 s_0^2 \Big(-\Omega + \frac{U_0 a}{\Sigma_0} \Big)}{\Sigma_0} \\ -\Omega \Big(a^2 + r_0^2 \Big) \Bigg) \end{matrix} \frac{}{r_0^2} & \begin{matrix} 2c_0 s_0 \Bigg(\frac{2Ma^2 r_0 s_0^2 \Big(1 + \frac{a^2 s_0^2}{\Sigma_0} \Big)}{\Sigma_0} \\ + \frac{2Ma^2 r_0 s_0^2}{\Sigma_0} + a^2 \\ + r_0^2 \Bigg) \end{matrix} \frac{}{r_0^3} & 0 & 0 \\
0 & 0 & \frac{2a^2 c_0 s_0}{\Delta_0 r_0} & 0 \\
0 & 0 & 0 & \frac{2a^2 c_0 s_0}{r_0^3}
\end{pmatrix}$$

$$\hat{M} = R_0\, \delta y + P_0\, \delta z =$$

$$\begin{pmatrix}
\begin{matrix} 2c_0 \delta z s_0 \Bigg(\frac{4M\Omega U_0 a r_0}{\Sigma_0} \\ -\frac{2MU_0^2 a^2 r_0}{\Sigma_0^2} - \Omega^2 \Big(a^2 + r_0^2 \Big) \Bigg) \end{matrix} \frac{}{r_0} & \begin{matrix} 2c_0 \delta z s_0 \Bigg(\frac{2MU_0 a r_0}{\Sigma_0} \\ + \frac{2Ma^2 r_0 s_0^2 \Big(-\Omega + \frac{U_0 a}{\Sigma_0} \Big)}{\Sigma_0} \\ -\Omega \Big(a^2 + r_0^2 \Big) \Bigg) \end{matrix} \frac{}{r_0^2} & 0 & 0 \\
\begin{matrix} +2\delta y \Bigg(-\frac{MU_0^2}{\Sigma_0} + \frac{2MU_0^2 r_0^2}{\Sigma_0^2} \\ -\Omega^2 r_0 s_0^2 \Bigg) \end{matrix} & \begin{matrix} 2\delta y s_0^2 \Bigg(\frac{MU_0 a}{\Sigma_0} \\ + \frac{-\frac{2MU_0 a r_0^2}{\Sigma_0^2} - \Omega r_0 \Big)}{r_0} \end{matrix} & & \\
\begin{matrix} 2c_0 \delta z s_0 \Bigg(\frac{2MU_0 a r_0}{\Sigma_0} \\ + \frac{2Ma^2 r_0 s_0^2 \Big(-\Omega + \frac{U_0 a}{\Sigma_0} \Big)}{\Sigma_0} \\ -\Omega \Big(a^2 + r_0^2 \Big) \Bigg) \end{matrix} \frac{}{r_0^2} & \begin{matrix} 2c_0 \delta z s_0 \Bigg(\frac{2Ma^2 r_0 s_0^2 \Big(1 + \frac{a^2 s_0^2}{\Sigma_0} \Big)}{\Sigma_0} \\ + \frac{2Ma^2 r_0 s_0^2}{\Sigma_0} + a^2 + r_0^2 \\ \Bigg) \end{matrix} \frac{}{r_0^3} & 0 & 0 \\
\begin{matrix} 2\delta y s_0^2 \Bigg(\frac{MU_0 a}{\Sigma_0} \\ + \frac{-\frac{2MU_0 a r_0^2}{\Sigma_0^2} - \Omega r_0 \Big)}{r_0} \end{matrix} & \begin{matrix} 2\delta y s_0^2 \Bigg(\frac{Ma^2 s_0^2}{\Sigma_0} \\ - \frac{-\frac{2Ma^2 r_0^2 s_0^2}{\Sigma_0^2} + r_0 \Big)}{r_0^2} \end{matrix} & & \\
0 & 0 & \frac{2a^2 c_0 \delta z s_0}{\Delta_0 r_0} & 0 \\
0 & 0 & -\frac{2\delta y \Big(r_0 + \frac{\Sigma_0 \Big(M - r_0 \Big)}{\Delta_0} \Big)}{\Delta_0} & \frac{2a^2 c_0 \delta z s_0}{r_0^3} - \frac{2\delta y}{r_0}
\end{pmatrix}.$$

When $\theta_0 = \pi/2$,

$$\hat{M}\Big|_{\theta_0=\pi/2} = \delta y \begin{pmatrix} 2\left(\frac{M(-\Omega a+1)^2}{r_0^2} - \Omega^2 r_0\right) & -\frac{2\left(\frac{Ma(-\Omega a+1)}{r_0^2}+\Omega r_0\right)}{r_0} & 0 & 0 \\ -\frac{2\left(\frac{Ma(-\Omega a+1)}{r_0^2}+\Omega r_0\right)}{r_0} & -\frac{2\left(-\frac{Ma^2}{r_0^2}+r_0\right)}{r_0^2} & 0 & 0 \\ 0 & 0 & -\frac{2r_0\left(1+\frac{r_0(M-r_0)}{\Delta_0}\right)}{\Delta_0} & 0 \\ 0 & 0 & 0 & -\frac{2}{r_0} \end{pmatrix}$$

(This is because, when $\theta_0 = \pi/2$, the factor $c_0 = \cos\theta_0$ appearing in every term of $P_0 = \partial_z \hat{g}|_0$ is 0.) The center of the cavity has fixed spatial coordinates in the frame at rest relative to the cavity. Thus, along its central trajectory,

$$dx = dy = dz = 0$$

and

$$ds^2 = C^{-2}\left(\Omega; r_0, \theta_0\right)\big|_{\theta_0=\pi/2}\, dt^2$$

$$\begin{aligned} C^{-2}\left(\Omega; r, \theta\right) &= g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi} \\ &= \left(1 - \frac{2Mr}{\Sigma}\right) + 2\Omega\frac{2Mar}{\Sigma}\sin^2\theta + \Omega^2\left[-\left(r^2 + a^2 + \frac{2Mra^2\sin^2\theta}{\Sigma}\right)\sin^2\theta\right] \\ &= 1 - \frac{2Mr}{\Sigma} + \frac{2Mr}{\Sigma}\left(2a\Omega\sin^2\theta\right) - \Omega^2\left(r^2 + a^2\right)\sin^2\theta - \frac{2Mr}{\Sigma}\left(\Omega^2 a^2\sin^4\theta\right) \\ &= 1 - \Omega^2\left(r^2 + a^2\right)\sin^2\theta - \frac{2Mr}{\Sigma}\left(1 - 2a\Omega\sin^2\theta + \Omega^2 a^2\sin^4\theta\right) \end{aligned}$$

$$\begin{aligned} g_{\phi\phi} = -\frac{A}{\Sigma}\sin^2\theta &= -\frac{\left(r^2 + a^2\right)\Sigma + 2Mra^2\sin^2\theta}{\Sigma}\sin^2\theta \\ &= -\left(r^2 + a^2 + \frac{2Mra^2\sin^2\theta}{\Sigma}\right)\sin^2\theta \end{aligned}$$

where we have used

$$A = \left(r^2 + a^2\right)\Sigma + 2Mra^2\sin^2\theta$$

Taking the cavity center to be at $r_0, \theta_0 = \pi/2$, we obtain

$$\begin{aligned} C^{-2}\left(\Omega; r_0, \theta_0\right)\big|_{\theta_0=\pi/2} &= 1 - \Omega^2\left(r_0^2 + a^2\right) - \frac{2Mr_0}{\Sigma|_{\theta_0=\pi/2}}\left(1 - 2a\Omega + \Omega^2 a^2\right) \\ &= 1 - \Omega^2\left(r_0^2 + a^2\right) - \frac{2M}{r_0}\left(1 - a\Omega\right)^2 \end{aligned}$$

where

$$\Sigma|_{\theta_0=\pi/2} = r_0^2 + a^2\cos^2\left(\frac{\pi}{2}\right) = r_0^2$$

Define

$$u = \frac{M}{r_0}, j = \frac{a}{r_0}, v = \Omega r_0, d_0 = \frac{\Delta_0}{r_0^2}$$

where

$$\Delta_0 = r_0^2 - 2Mr_0 + a^2$$

Then

$$\begin{aligned} C^{-2}\left(\Omega; r_0, \theta_0\right)\big|_{\theta_0=\pi/2} &= 1 - \Omega^2\left(r_0^2 + a^2\right) - \frac{2M}{r_0}\left(1 - a\Omega\right)^2 \\ &= 1 - \left(1 + \frac{a^2}{r_0^2}\right)\Omega^2 r_0^2 - 2\frac{M}{r_0}\left(1 - \frac{a}{r_0}\cdot\Omega r_0\right)^2 \\ &= 1 - \left(1 + j^2\right)v^2 - 2u\left(1 - jv\right)^2 \end{aligned}$$

A massive observer must follow a timelike trajectory, so

$$ds^2 = C^{-2}\left(\Omega; r_0, \theta_0\right)\big|_{\theta_0=\pi/2}\, dt^2 > 0$$

Hence,

$$C^{-2}\left(\Omega; r_0, \theta_0\right)\big|_{\theta_0=\pi/2} = 1 - \left(1+j^2\right)v^2 - 2u\left(1-jv\right)^2 > 0$$

which gives

$$\left(1+j^2\right)v^2 + 2u\left(1-jv\right)^2 < 1$$

Since $u > 0, \left(1-jv\right)^2 \geq 0$, we obtain

$$\left(1+j^2\right)v^2 \leq \left(1+j^2\right)v^2 + 2u\left(1-jv\right)^2 < 1$$

Using $1 + j^2 \geq 1$ also gives

$$v^2 < \frac{1}{1+j^2} \leq 1$$

and hence

$$|v| < \frac{1}{\sqrt{1+j^2}} \leq 1$$

Thus,

$$|v| \leq 1,$$

or equivalently,

$$|\Omega r_0| \leq 1.$$

From

$$r_0 > r_+ \geq M > 0$$

we have

$$0 < \frac{M}{r_0} < 1$$

From

$$|a| \leq M, r_0 > 0$$

we obtain

$$\frac{|a|}{r_0} \leq \frac{M}{r_0} < 1.$$

Using

$$\begin{aligned} d_0 = \frac{\Delta_0}{r_0^2} &= \frac{r_0^2 - 2Mr_0 + a^2}{r_0^2} \\ &= 1 - \frac{2M}{r_0} + \frac{a^2}{r_0^2} \end{aligned}$$

and

$$\frac{a^2}{r_0^2} \leq \frac{M^2}{r_0^2} < \frac{M}{r_0}$$

and recalling that $0 < M/r_0 < 1$, we obtain

$$d_0 = 1 - \frac{2M}{r_0} + \frac{a^2}{r_0^2} < 1 - \frac{M}{r_0} < 1$$

Outside the outer event horizon,

$$\Delta_0 > 0$$

Therefore,

$$0 < d_0 < 1$$

(The outer event horizon of a Kerr black hole is $r_+ = M + \sqrt{M^2 - a^2}$. Requiring the radicand to be nonnegative gives the maximum magnitude of the specific angular momentum $a = J/M$ as $|a|_{\max} = M$, in which case the outer event horizon is $r_+ = M$. When $a = 0$, we have $r_+ = 2M$.)

Let $L_{\mathcal{C}}$ denote the maximum extent of the entire finite parallel-plate cavity $\mathcal{C}$, including both the plate separation and the finite transverse dimensions. Define

$$L_{\mathcal{C}} = \sup_{\boldsymbol{x} \in \mathcal{C}} \max_i |\delta x^i|,$$

and

$$\varepsilon = \frac{L_{\mathcal{C}}}{r_0}.$$

Writing $U_0 = 1 - a\Omega$, we have

$$\hat{M}\Big|_{\theta_0=\pi/2} = \delta y\, R_0 = \delta y \begin{pmatrix} 2\frac{MU_0^2}{r_0^2} - 2\Omega^2 r_0 & -2\frac{MaU_0}{r_0^3} - 2\Omega & 0 & 0 \\ -2\frac{MaU_0}{r_0^3} - 2\Omega & 2\frac{Ma^2}{r_0^4} - 2\frac{1}{r_0} & 0 & 0 \\ 0 & 0 & \frac{2r_0\left(Mr_0 - a^2\right)}{\Delta_0^2} & 0 \\ 0 & 0 & 0 & -\frac{2}{r_0} \end{pmatrix}$$

where

$$\begin{aligned} -\frac{2r_0\left(1 + \frac{r_0(M-r_0)}{\Delta_0}\right)}{\Delta_0} &= -\frac{2r_0\left[\Delta_0 + r_0\left(M - r_0\right)\right]}{\Delta_0^2} \\ &= -\frac{2r_0\left[\left(r_0^2 - 2Mr_0 + a^2\right) + r_0\left(M - r_0\right)\right]}{\Delta_0^2} \\ &= \frac{2r_0\left(Mr_0 - a^2\right)}{\Delta_0^2} \end{aligned}$$

$$|U_0| = |1 - a\Omega| \le 1 + |a\Omega| = 1 + \frac{|a|}{r_0}|\Omega| r_0 < 2$$

(Here we have used the triangle inequality $|x + y| \le |x| + |y|$ and the previously established bounds $|\Omega r_0| \le 1$ and $|a|/r_0 < 1$.) Under the stated convention, $|U_0| < 2$ also implies $|U_0| \le 2$. Thus,

$$r_0\left|(R_0)_{00}\right| = \left|2\frac{MU_0^2}{r_0} - 2\Omega^2 r_0^2\right| \le \left|2\frac{M}{r_0}U_0^2\right| + \left|2\Omega^2 r_0^2\right| < 8 + 2 = 10,$$

$$r_0\left|(R_0)_{01}\right| = r_0\left|(R_0)_{10}\right| = \left|-2\frac{MaU_0}{r_0^2} - 2\Omega r_0\right| \le \left|2\frac{M}{r_0}\frac{a}{r_0}U_0\right| + \left|2\Omega r_0\right| < 4 + 2 = 6,$$

Using

$$0 \le \frac{M}{r_0}\frac{a^2}{r_0^2} < 1 \Rightarrow -2 \le 2\frac{M}{r_0}\frac{a^2}{r_0^2} - 2 < 0 \Rightarrow 2 \ge 2 - 2\frac{M}{r_0}\frac{a^2}{r_0^2} = \left|2\frac{Ma^2}{r_0^3} - 2\right| > 0$$

gives

$$r_0\left|(R_0)_{11}\right| = \left|2\frac{Ma^2}{r_0^3} - 2\right| \le 2,$$

$$r_0\left|(R_0)_{33}\right| = 2,$$

We obtain

$$\left|\hat{M}_{00}\right| = \frac{r_0\left|(R_0)_{00}\right|}{r_0}\,|\delta y| < \frac{10}{r_0}\cdot L_{\mathcal{C}} = 10\varepsilon \le 10\varepsilon$$

(Under the stated convention, $\left|\hat{M}_{00}\right| < 10\varepsilon$ naturally implies $\left|\hat{M}_{00}\right| \le 10\varepsilon$.) Similarly,

$$\left|\hat{M}_{01}\right| = \left|\hat{M}_{10}\right| \le 6\varepsilon,$$

$$\left|\hat{M}_{11}\right| \le 2\varepsilon,$$

$$\left|\hat{M}_{33}\right| = \frac{r_0\left|(R_0)_{33}\right|}{r_0}\,|\delta y| \le \frac{r_0\left|(R_0)_{33}\right|}{r_0}\cdot L_{\mathcal{C}} = 2\varepsilon.$$

while

$$\left|\hat{M}_{22}\right| = \left|(R_0)_{22}\right|\,|\delta y| \le \frac{2r_0\left(Mr_0 - a^2\right)}{\Delta_0^2}\cdot L_{\mathcal{C}} \le \frac{2Mr_0^2}{\Delta_0^2}L_{\mathcal{C}} = \frac{2M}{r_0}\frac{L_{\mathcal{C}}/r_0}{\left(\Delta_0/r_0^2\right)^2} \le \frac{2\varepsilon}{d_0^2}.$$

Since $0 < d_0 < 1$, we have $1/d_0^2 > 1$, and hence

$$\left|\hat{M}_{00}\right| \le 10\varepsilon < \frac{10\varepsilon}{d_0^2} \le \frac{10\varepsilon}{d_0^2},$$

$$\left|\hat{M}_{01}\right| = \left|\hat{M}_{10}\right| < \frac{6\varepsilon}{d_0^2} \le \frac{6\varepsilon}{d_0^2},$$

$$\left|\hat{M}_{11}\right| < \frac{2\varepsilon}{d_0^2} \le \frac{2\varepsilon}{d_0^2},$$

$$\left|\hat{M}_{33}\right| \leq 2\varepsilon < \frac{2\varepsilon}{d_0^2} \leq \frac{2\varepsilon}{d_0^2}.$$

This gives

$$\max_{\mu,\nu} \left|\hat{M}_{\mu\nu}\right| \leq \frac{10\varepsilon}{d_0^2}.$$

By Taylor's theorem with the mean-value form of the remainder, $\hat{g}_{\mu\nu}\left(t,x,y,z\right)$ can be written as

$$\hat{g}_{\mu\nu}\left(t,x,y,z\right) = \hat{g}_{\mu\nu}\left(t,x,y,z\right)\big|_{\boldsymbol{x}=\boldsymbol{x}_0} + \sum_{i=1}^{3} \left.\frac{\partial \hat{g}_{\mu\nu}\left(t,x,y,z\right)}{\partial x^i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0} \left(x^i - x_0^i\right) + \mathcal{R}_{\mu\nu}$$

where

$$\mathcal{R}_{\mu\nu} = \frac{1}{2}\sum_{i,j=1}^{3} \left.\frac{\partial^2 \hat{g}_{\mu\nu}\left(t,x,y,z\right)}{\partial x^i \partial x^j}\right|_{\boldsymbol{x}=\boldsymbol{\xi}} \left(x^i - x_0^i\right)\left(x^j - x_0^j\right)$$

Here, $\boldsymbol{\xi} = (\xi_1, \xi_2, \xi_3)$, and $\xi_1,\ \xi_2,\ \xi_3$ lie between x_0 and x, y_0 and y, and z_0 and z, respectively (or, equivalently, $\boldsymbol{\xi} = \boldsymbol{x}_0 + s\left(\boldsymbol{x} - \boldsymbol{x}_0\right),\ 0 < s < 1$). Let

$$\bar{K}(\boldsymbol{\xi}) = \frac{1}{2}\sum_{i,j=1}^{3} \max_{\mu,\nu} \left|\left.\frac{\partial^2 \hat{g}_{\mu\nu}\left(t,x,y,z\right)}{\partial x^i \partial x^j}\right|_{\boldsymbol{x}=\boldsymbol{\xi}}\right|,$$

To specify the range of $\boldsymbol{\xi}$ over which $\bar{K}(\boldsymbol{\xi})$ attains its maximum, first fix a small closed neighborhood $\mathcal{N}$ outside the outer event horizon, and then consider a small cavity contained in it. Let

$$\tilde{K} = \frac{1}{2}\sum_{i,j=1}^{3} \max_{\boldsymbol{\xi}\in\mathcal{N};\mu,\nu} \left|\left.\frac{\partial^2 \hat{g}_{\mu\nu}\left(t,x,y,z\right)}{\partial x^i \partial x^j}\right|_{\boldsymbol{x}=\boldsymbol{\xi}}\right|,\ K = \tilde{K} r_0^2$$

Since

$$\left|x^i - x_0^i\right| \leq L_{\mathcal{C}}$$

every component $|\mathcal{R}_{\mu\nu}|$, including the largest one, satisfies

$$|\mathcal{R}_{\mu\nu}| \leq \frac{1}{2}\sum_{i,j=1}^{3} \left|\left.\frac{\partial^2 \hat{g}_{\mu\nu}\left(t,x,y,z\right)}{\partial x^i \partial x^j}\right|_{\boldsymbol{x}=\boldsymbol{\xi}}\right| \left|x^i - x_0^i\right| \left|x^j - x_0^j\right| \leq \tilde{K} L_{\mathcal{C}}^2 = \tilde{K} r_0^2 \frac{L_{\mathcal{C}}^2}{r_0^2} = K\varepsilon^2$$

The first $\leq$ follows because the absolute value of a sum does not exceed the sum of the absolute values. Thus,

$$\max_{\mu,\nu} |\mathcal{R}_{\mu\nu}| \leq K\varepsilon^2$$

Consequently, for all μ, ν,

$$\begin{aligned}
\left|\hat{g}_{\mu\nu}\left(t,x,y,z\right) - \hat{g}_{\mu\nu}\left(t,x,y,z\right)\big|_{\boldsymbol{x}=\boldsymbol{x}_0}\right| &= \left|\hat{M}_{\mu\nu} + \mathcal{R}_{\mu\nu}\right| \\
&\leq \left|\hat{M}_{\mu\nu}\right| + |\mathcal{R}_{\mu\nu}| \\
&\leq \max_{\mu,\nu}\left|\hat{M}_{\mu\nu}\right| + \max_{\mu,\nu}|\mathcal{R}_{\mu\nu}| \\
&\leq \frac{10\varepsilon}{d_0^2} + K\varepsilon^2
\end{aligned}$$

For a fixed orbit (d_0 and K are finite constants) and sufficiently small ε, we therefore have

$$\max_{\mu,\nu}\left|\hat{g}_{\mu\nu}\left(t,x,y,z\right) - \hat{g}_{\mu\nu}\left(t,x,y,z\right)\big|_{\boldsymbol{x}=\boldsymbol{x}_0}\right| \leq \frac{10\varepsilon}{d_0^2} + K\varepsilon^2 \ll 1.$$

In other words, the maximum variation of the metric components near $\boldsymbol{x}_0$ satisfies

$$\max_{\mu,\nu}|\delta\hat{g}_{\mu\nu}| = \max_{\mu,\nu}\left|\hat{g}_{\mu\nu}\left(t,x,y,z\right) - \hat{g}_{\mu\nu}\left(t,x,y,z\right)\big|_{\boldsymbol{x}=\boldsymbol{x}_0}\right| \ll 1$$

, so the metric components can be treated as constants throughout the region enclosed by the parallel plates. The ratio of the bound on the higher-order remainder to the bound on the first-order term is

$$P = \frac{K\varepsilon^2}{10\varepsilon/d_0^2} = \frac{Kd_0^2}{10}\varepsilon = C\varepsilon,$$

where C is a finite constant. As $\varepsilon \to 0$, the bound on the higher-order remainder decreases faster than the bound on the first-order term. We now make a numerical estimate using the first-order approximation. Let $M_{\rm BH}$ be the black hole mass in SI units and M the black hole mass in geometrized units. They are related by

$$M = \frac{GM_{\rm BH}}{c^2},$$

For a black hole of one solar mass,

$$M = \frac{GM_\odot}{c^2} = \frac{6.674 \times 10^{-11}\ \mathrm{m^3 kg^{-1} s^{-2}} \times 1.988 \times 10^{30}\ \mathrm{kg}}{(299792458\ \mathrm{m/s})^2} \approx 1.476\ \mathrm{km},$$

When the black hole mass is 6.775 solar masses,

$$M \approx 10\ \mathrm{km}.$$

1. Taking $a = 0.998M$, $r_0 = 4M = 40$ km, and $L_{\mathcal{C}} = 1$ mm, we have $d_0 = 1 - (2M/r_0) + \left(a^2/r_0^2\right) \approx 0.562$ and $\varepsilon = L_{\mathcal{C}}/r_0 = 10^{-3}/\left(4 \times 10^4\right) = 2.5 \times 10^{-8}$. Hence,

$$\max_{\mu,\nu} |\delta \hat{g}^{(1)}_{\mu\nu}| \leq \frac{10\varepsilon}{d_0^2} \approx 7.915 \times 10^{-7} \ll 1,$$

The superscript (1) denotes the first-order approximation to the metric components. 2. Taking $a = 0.998M$, $r_0 = 1.5M = 15$ km, and $L_{\mathcal{C}} = 1$ mm, we have $d_0 = 1 - (2M/r_0) + \left(a^2/r_0^2\right) \approx 0.109$ and $\varepsilon = L_{\mathcal{C}}/r_0 \approx 6.667 \times 10^{-8}$. Hence,

$$\max_{\mu,\nu} |\delta \hat{g}^{(1)}_{\mu\nu}| \leq \frac{10\varepsilon}{d_0^2} \approx 5.611 \times 10^{-5} \ll 1.$$

Thus, in this physical setting, with a fixed orbit and a sufficiently small cavity, the metric components vary very little near $\boldsymbol{x}_0$ in the first-order approximation. Since the bound on the higher-order remainder decreases faster than that on the first-order term as $\varepsilon \to 0$, the components of the unapproximated metric also vary very little near $\boldsymbol{x}_0$. The metric components can therefore reasonably be approximated as constants, using the zeroth-order approximation with small corrections.

III. The Klein-Gordon Equation and Quantum Field Mode Solutions

The Klein-Gordon equation for a scalar field in curved spacetime is given in Spacetime and Geometry: An Introduction to General Relativity by Sean M. Carroll, published by Cambridge University Press. With the metric signature $(-,+,+,+)$, which is opposite to that used in the present paper, it reads

$$\Box\Psi - m^2\Psi - \xi R\Psi = 0$$

For a massless scalar field in vacuum, the equation becomes

$\Box\Psi = 0$

$\nabla_\mu \nabla^\mu \Psi = 0$

$\nabla_\mu \left(g^{\mu\nu} \partial_\nu \Psi \right) = 0$

For a scalar field Ψ, the covariant derivative ∇_μ and the ordinary derivative ∂_μ coincide

$\partial_\mu \left(g^{\mu\nu} \partial_\nu \Psi \right) + \Gamma^\mu_{\mu\alpha} \left(g^{\alpha\nu} \partial_\nu \Psi \right) = 0$

For a vector field $V^\mu = g^{\mu\nu}\partial_\nu\Psi$, the covariant derivative $\nabla_\mu V^\mu = \partial_\mu V^\mu + \Gamma^\mu_{\mu\alpha} V^\alpha$

$\partial_\mu \left(g^{\mu\nu} \partial_\nu \Psi \right) + \frac{1}{\sqrt{-g}} \partial_\alpha \left(\sqrt{-g} \right) \left(g^{\alpha\nu} \partial_\nu \Psi \right) = 0$

It can be shown that $\Gamma^\mu_{\mu\alpha} = \frac{1}{\sqrt{-g}} \partial_\alpha \left(\sqrt{-g} \right)$; g=det$\left(g_{\mu\nu} \right)$

$\frac{1}{\sqrt{-g}} \left[\sqrt{-g} \partial_\mu \left(g^{\mu\nu} \partial_\nu \Psi \right) + \partial_\mu \left(\sqrt{-g} \right) \left(g^{\mu\nu} \partial_\nu \Psi \right) \right] = 0$

$\frac{1}{\sqrt{-g}} \left[\partial_\mu \left(\sqrt{-g} g^{\mu\nu} \partial_\nu \Psi \right) \right] = 0$

product rule for differentiation

Since the metric matrix is treated as constant, the factor $\sqrt{-\hat{g}}\hat{g}^{\mu\nu}$ (with $\sqrt{-\hat{g}}=1$) can be taken outside the derivative: $\frac{1}{\sqrt{-\hat{g}}}\left[\partial_\mu\left(\sqrt{-\hat{g}}\hat{g}^{\mu\nu}\partial_\nu\Psi\right)\right]=\frac{\sqrt{-\hat{g}}}{\sqrt{-\hat{g}}}\hat{g}^{\mu\nu}\partial_\mu\partial_\nu\Psi$. Thus, we obtain the following equation. Here, Ψ represents only one mode of the field, corresponding to a discrete frequency ω_n and a continuous wave vector $\vec{k}_\perp$, rather than the full field.

$$\begin{aligned}
\hat{g}^{\mu\nu}\partial_\mu\partial_\nu\Psi&=0\\
\left(\frac{A}{\Delta r^2}\partial_t^2+2\frac{A}{\Delta r}\left(\omega_d-\Omega\right)\partial_t\partial_x-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\partial_x^2-\frac{\Delta}{r^2}\partial_y^2-\partial_z^2\right)e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\chi\left(x\right)&=0\\
\left[\frac{A}{\Delta r^2}\left(-\omega_n^2\right)+2\frac{A}{\Delta r}\left(\omega_d-\Omega\right)\left(-i\omega_n\right)\partial_x-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\partial_x^2-\frac{\Delta}{r^2}\left(-k_y^2\right)-\left(-k_z^2\right)\right]e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\chi\left(x\right)&=0\\
e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\partial_x^2\chi+2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\partial_x\chi+\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right)\chi\right]&=0
\end{aligned}$$

where

$$\begin{aligned}
\partial_t^2\Psi&=\partial_t^2e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\chi\left(x\right)\\
&=\left(-i\omega_n\right)^2e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\chi\left(x\right)\\
&=-\omega_n^2e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\chi\left(x\right)\\
&=-\omega_n^2\Psi
\end{aligned}$$

$$\begin{aligned}
\partial_t\partial_x\Psi&=\partial_t\partial_xe^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\chi\left(x\right)\\
&=\left(-i\omega_n\right)e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\partial_x\chi\left(x\right)\\
&=\left(-i\omega_n\right)\partial_x\Psi
\end{aligned}$$

$$\begin{aligned}
\partial_y^2\Psi&=\partial_y^2e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}\chi\left(x\right)\\
&=\partial_y^2e^{-i\omega_n t}e^{i\left(k_y y+k_z z\right)}\chi\left(x\right)\\
&=\left(ik_y\right)^2e^{-i\omega_n t}e^{i\left(k_y y+k_z z\right)}\chi\left(x\right)\\
&=-k_y^2\Psi
\end{aligned}$$

Similarly,

$$\partial_z^2\Psi=-k_z^2\Psi$$

Seek a solution of

$$-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\partial_x^2\chi+2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\partial_x\chi+\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right)\chi=0$$

in the form

$$\chi=e^{\lambda x}$$

(corresponding to $\Psi=e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}e^{\lambda x}$). Then

$$\begin{aligned}
-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\partial_x^2e^{\lambda x}+2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\partial_xe^{\lambda x}+\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right)e^{\lambda x}&=0\\
\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\lambda^2+2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\lambda+\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right)\right]e^{\lambda x}&=0
\end{aligned}$$

This requires the characteristic equation

$$-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\lambda^2+2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\lambda-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2=0$$

For the quadratic equation $a\lambda^2+b\lambda+c=0$, the discriminant is

$$\begin{aligned}
\tilde{\Delta} &= b^2-4ac \\
&= \left[2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]^2-4\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right]\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= \left[2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]^2+4\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= -\left[2\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]^2+4\frac{r^2}{\Delta}\left\{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2\right]\right\}\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= -\left[2\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]^2+4\left[\frac{r^4}{A}-\frac{A}{\Delta}\left(\Omega-\omega_d\right)^2\right]\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= -\left[2\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]^2+4\frac{r^4}{A}\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right)+\left[2\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]^2-4\frac{A}{\Delta}\left(\Omega-\omega_d\right)^2\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= 4\frac{r^4}{A}\left(-\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2\right)-4\frac{A}{\Delta}\left(\Omega-\omega_d\right)^2\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= 4\frac{r^4}{A}\left(-\frac{A}{\Delta r^2}\omega_n^2\right)+4\frac{r^4}{A}\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)-4\frac{A}{\Delta}\left(\Omega-\omega_d\right)^2\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= -4\frac{r^2}{\Delta}\omega_n^2+4\left[\frac{r^4}{A}-\frac{A}{\Delta}\left(\Omega-\omega_d\right)^2\right]\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= -4\frac{r^2}{\Delta}\omega_n^2+4\frac{r^4}{A}\left[1-\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2\right]\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= -4\frac{r^2}{\Delta}\omega_n^2+4\frac{r^2}{\Delta}\left\{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2\right]\right\}\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= -4\frac{r^2}{\Delta}\omega_n^2+4\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)
\end{aligned}$$

The roots of the characteristic equation are

$$\begin{aligned}
\lambda_{1,2} &= \frac{-b\pm\sqrt{\tilde{\Delta}}}{2a} \\
&= \frac{-b\pm\sqrt{i^2\cdot i^{-2}\tilde{\Delta}}}{2a} \\
&= \frac{-b\pm i\sqrt{\left(-1\right)\tilde{\Delta}}}{2a} \\
&= \frac{-b}{2a}\pm i\frac{\sqrt{-\tilde{\Delta}}}{2a} \\
&= \frac{-\left[2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]}{2\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right]}\pm i\frac{\sqrt{-\left[-4\frac{r^2}{\Delta}\omega_n^2+4\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\right]}}{2\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right]}
\end{aligned}$$

Let

$$\begin{aligned}
\tilde{\alpha} &= -\frac{b}{2a} \\
&= \frac{-\left[2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)\right]}{2\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right]} \\
&= i\frac{\omega_n A\left(\Omega-\omega_d\right)}{r^3C^{-2}\left(\Omega\right)} \\
&= -i\frac{\omega_n A\left(\omega_d-\Omega\right)}{r^3C^{-2}\left(\Omega\right)}
\end{aligned}$$

$$\begin{aligned}
\tilde{\beta} &= \frac{\sqrt{-\tilde{\Delta}}}{2a} \\
&= \frac{\sqrt{-\left[-4\frac{r^2}{\Delta}\omega_n^2+4\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\right]}}{2\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right]}
\end{aligned}$$

Then $\lambda_{1,2} = \tilde{\alpha} \pm i\tilde{\beta}$, and the general solution is

$$\begin{aligned}
\chi &= \tilde{C}_1 e^{\lambda_1 x} + \tilde{C}_2 e^{\lambda_2 x} \\
&= \tilde{C}_1 e^{\left(\tilde{\alpha}+i\tilde{\beta}\right)x} + \tilde{C}_2 e^{\left(\tilde{\alpha}-i\tilde{\beta}\right)x} \\
&= \tilde{C}_1 e^{\tilde{\alpha}x} e^{i\tilde{\beta}x} + \tilde{C}_2 e^{\tilde{\alpha}x} e^{-i\tilde{\beta}x} \\
&= e^{\tilde{\alpha}x}\left[\tilde{C}_1\left(\cos\tilde{\beta}x + i\sin\tilde{\beta}x\right) + \tilde{C}_2\left(\cos\tilde{\beta}x - i\sin\tilde{\beta}x\right)\right] \\
&= e^{\tilde{\alpha}x}\left[\left(\tilde{C}_1 + \tilde{C}_2\right)\cos\tilde{\beta}x + i\left(\tilde{C}_1 - \tilde{C}_2\right)\sin\tilde{\beta}x\right] \\
&= e^{\tilde{\alpha}x}\left[C_1\cos\tilde{\beta}x + C_2\sin\tilde{\beta}x\right]
\end{aligned}$$

where $\tilde{C}_1$ and $\tilde{C}_2$ are arbitrary complex constants. Define $C_1 = \tilde{C}_1 + \tilde{C}_2$ and $C_2 = i\left(\tilde{C}_1 - \tilde{C}_2\right)$. We may choose $\tilde{C}_1 = \left(C_1 - iC_2\right)/2$ and $\tilde{C}_2 = \left(C_1 + iC_2\right)/2$ so that C_1 and C_2 are arbitrary real constants. The Dirichlet boundary conditions

$$\Psi\left(t, x = 0, y, z\right) = \Psi\left(t, x = L, y, z\right) = 0$$

give

$$\begin{aligned}
\chi\left(x = 0\right) &= e^{\tilde{\alpha}\cdot 0}\left(C_1\cos\tilde{\beta}\cdot 0 + C_2\sin\tilde{\beta}\cdot 0\right) \\
&= C_1 = 0
\end{aligned}$$

$$\begin{aligned}
\chi\left(x = L\right) &= e^{\tilde{\alpha}\cdot L}\left(0\cdot\cos\tilde{\beta}\cdot L + C_2\sin\tilde{\beta}\cdot L\right) \\
&= C_2 e^{-i\frac{\omega_n A\left(\omega_d-\Omega\right)}{r^3 C^{-2}(\Omega)}\cdot L}\sin\tilde{\beta}L \\
&= 0
\end{aligned}$$

To obtain a nontrivial solution, that is, a solution that is not identically zero, we require

$$\sin\tilde{\beta}L = 0$$

Hence,

$$\begin{aligned}
\tilde{\beta}L &= n\pi \\
\Rightarrow\tilde{\beta} &= \frac{n\pi}{L}
\end{aligned}$$

and we obtain

$$\chi = C_2 e^{-i\frac{\omega_n A\left(\omega_d-\Omega\right)}{r^3 C^{-2}(\Omega)}x}\sin\left(\frac{n\pi}{L}x\right)$$

Method 2:

$$\gamma_2\partial_x^2\chi + \gamma_1\partial_x\chi + \gamma_0\chi = 0$$

where

$$\begin{aligned}
\gamma_2 &= -\frac{r^2}{\Delta}C^{-2}\left(\Omega\right) \\
\gamma_1 &= 2i\omega_n\frac{A}{\Delta r}\left(\Omega - \omega_d\right) \\
\gamma_0 &= -\frac{A}{\Delta r^2}\omega_n^2 + \frac{\Delta}{r^2}k_y^2 + k_z^2
\end{aligned}$$

Let

$$\chi = e^{-i\beta x}\tilde{\chi}$$

Then

$$\begin{aligned}
\partial_x\chi &= \left(-i\beta\right)e^{-i\beta x}\tilde{\chi} + e^{-i\beta x}\partial_x\tilde{\chi} \\
&= e^{-i\beta x}\left(-i\beta\tilde{\chi} + \partial_x\tilde{\chi}\right)
\end{aligned}$$

$$\begin{aligned}
\partial_x^2\chi &= \partial_x\left[e^{-i\beta x}\left(-i\beta\tilde{\chi} + \partial_x\tilde{\chi}\right)\right] \\
&= \left(-i\beta\right)e^{-i\beta x}\left(-i\beta\tilde{\chi} + \partial_x\tilde{\chi}\right) + e^{-i\beta x}\left(-i\beta\partial_x\tilde{\chi} + \partial_x^2\tilde{\chi}\right) \\
&= e^{-i\beta x}\left[-\beta^2\tilde{\chi} - i2\beta\partial_x\tilde{\chi} + \partial_x^2\tilde{\chi}\right]
\end{aligned}$$

Substitution into the original equation gives

$$\begin{aligned}\gamma_2 e^{-i\beta x}\left[-\beta^2\tilde{\chi}-i2\beta\partial_x\tilde{\chi}+\partial_x^2\tilde{\chi}\right]+\gamma_1 e^{-i\beta x}\left(-i\beta\tilde{\chi}+\partial_x\tilde{\chi}\right)+\gamma_0 e^{-i\beta x}\tilde{\chi}=0\\ \gamma_2\left[-\beta^2\tilde{\chi}-i2\beta\partial_x\tilde{\chi}+\partial_x^2\tilde{\chi}\right]+\gamma_1\left(-i\beta\tilde{\chi}+\partial_x\tilde{\chi}\right)+\gamma_0\tilde{\chi}=0\\ \gamma_2\partial_x^2\tilde{\chi}+\left(\gamma_1-i2\beta\gamma_2\right)\partial_x\tilde{\chi}+\left(\gamma_0-\beta^2\gamma_2-i\beta\gamma_1\right)\tilde{\chi}=0\end{aligned}$$

Set

$$\gamma_1-i2\beta\gamma_2=0$$

Then

$$\begin{aligned}\beta&=-\frac{\gamma_1}{2\gamma_2}i\\&=-\frac{2i\omega_n\frac{A}{\Delta r}\left(\Omega-\omega_d\right)}{2\left(-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right)}i\\&=\frac{\omega_n C^2\left(\Omega\right)A}{r^3}\left(\omega_d-\Omega\right)\\&=\frac{b_n A}{r^3}\left(\omega_d-\Omega\right)\end{aligned}$$

where we have defined

$$b_n=\omega_n C^2\left(\Omega\right)$$

We have

$$\begin{aligned}\gamma_2\partial_x^2\tilde{\chi}+\left(\gamma_0-\beta^2\gamma_2-i\beta\gamma_1\right)\tilde{\chi}=0\\ \gamma_2\partial_x^2\tilde{\chi}+\left[\gamma_0-\beta^2\gamma_2-i\beta\left(i2\beta\gamma_2\right)\right]\tilde{\chi}=0\\ \gamma_2\partial_x^2\tilde{\chi}+\left(\gamma_0+\beta^2\gamma_2\right)\tilde{\chi}=0\\ \partial_x^2\tilde{\chi}+\frac{\left(\gamma_0+\beta^2\gamma_2\right)}{\gamma_2}\tilde{\chi}=0\end{aligned}$$

Seek a solution of this equation in the form

$$\tilde{\chi}=e^{\lambda x}$$

(corresponding to $\Psi=e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}e^{-i\beta x}e^{\lambda x}$). Then

$$\left[\lambda^2+\frac{\left(\gamma_0+\beta^2\gamma_2\right)}{\gamma_2}\right]e^{\lambda x}=0$$

and hence

$$\lambda^2+\frac{\left(\gamma_0+\beta^2\gamma_2\right)}{\gamma_2}=0$$

$$\lambda_{1,2}=\pm\sqrt{-\frac{\left(\gamma_0+\beta^2\gamma_2\right)}{\gamma_2}}$$

Thus,

$$\begin{aligned}\chi&=e^{-i\beta x}\left(\tilde{C}_1e^{\lambda_1 x}+\tilde{C}_2e^{\lambda_2 x}\right)\\&=e^{-i\beta x}\left(\tilde{C}_1e^{\sqrt{-\frac{\left(\gamma_0+\beta^2\gamma_2\right)}{\gamma_2}}x}+\tilde{C}_2e^{-\sqrt{-\frac{\left(\gamma_0+\beta^2\gamma_2\right)}{\gamma_2}}x}\right)\\&=e^{-i\beta x}\left(\tilde{C}_1e^{i\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x}+\tilde{C}_2e^{-i\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x}\right)\\&=e^{-i\beta x}\left[\tilde{C}_1\left(\cos\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x+i\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x\right)+\tilde{C}_2\right.\\&\quad\left.\left(\cos\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x-i\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x\right)\right]\end{aligned}$$

$$= e^{-i\beta x}\left[\left(\tilde{C}_1+\tilde{C}_2\right)\cos\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x+i\left(\tilde{C}_1-\tilde{C}_2\right)\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x\right]$$

$$= e^{-i\frac{\omega_n A\left(\omega_d-\Omega\right)}{r^3C^{-2}\left(\Omega\right)}x}\left(C_1\cos\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x+C_2\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}x\right)$$

where

$$\begin{aligned}\gamma_0+\beta^2\gamma_2 &= -\frac{A}{\Delta r^2}\omega_n^2+\frac{\Delta}{r^2}k_y^2+k_z^2+\left[\frac{\omega_nC^2\left(\Omega\right)A}{r^3}\left(\omega_d-\Omega\right)\right]^2\left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right]\\ &= -\omega_n^2\left[\frac{A}{\Delta r^2}+\frac{C^2\left(\Omega\right)A^2}{r^6}\left(\omega_d-\Omega\right)^2\frac{r^2}{\Delta}\right]+\frac{\Delta}{r^2}k_y^2+k_z^2\\ &= -\omega_n^2C^2\left(\Omega\right)\left[C^{-2}\left(\Omega\right)\frac{A}{\Delta r^2}+\frac{A^2}{\Delta r^4}\left(\omega_d-\Omega\right)^2\right]+\frac{\Delta}{r^2}k_y^2+k_z^2\\ &= -\omega_n^2C^2\left(\Omega\right)\left(\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2\right]\frac{A}{\Delta r^2}+\frac{A^2}{\Delta r^4}\left(\omega_d-\Omega\right)^2\right)+\frac{\Delta}{r^2}k_y^2+k_z^2\\ &= -\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2\end{aligned}$$

The coefficients $\tilde{C}_1$ and $\tilde{C}_2$ are arbitrary complex constants. Define $C_1=\tilde{C}_1+\tilde{C}_2$ and $C_2=i\left(\tilde{C}_1-\tilde{C}_2\right)$. We may choose $\tilde{C}_1=\left(C_1-iC_2\right)/2$ and $\tilde{C}_2=\left(C_1+iC_2\right)/2$ so that C_1 and C_2 are arbitrary real constants. The Dirichlet boundary conditions

$$\Psi\left(t,x=0,y,z\right)=\Psi\left(t,x=L,y,z\right)=0$$

give

$$\begin{aligned}\chi\left(x=0\right) &= e^{-i\frac{\omega_nA\left(\omega_d-\Omega\right)}{r^3C^{-2}\left(\Omega\right)}\cdot0}\left(C_1\cos\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}\cdot0+C_2\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}\cdot0\right)\\ &= C_1=0\end{aligned}$$

$$\begin{aligned}\chi\left(x=L\right) &= e^{-i\frac{\omega_nA\left(\omega_d-\Omega\right)}{r^3C^{-2}\left(\Omega\right)}\cdot L}\left(0\cdot\cos\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}\cdot L+C_2\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}\cdot L\right)\\ &= C_2e^{-i\frac{\omega_nA\left(\omega_d-\Omega\right)}{r^3C^{-2}\left(\Omega\right)}L}\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}L\\ &= 0\end{aligned}$$

To obtain a nontrivial solution, that is, a solution that is not identically zero, we require

$$\sin\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}L=0$$

Hence,

$$\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}L=n\pi$$

$$\Rightarrow\sqrt{\frac{-\omega_n^2C^2\left(\Omega\right)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}}=\frac{n\pi}{L}$$

and we obtain

$$\chi=C_2e^{-i\frac{\omega_nA\left(\omega_d-\Omega\right)}{r^3C^{-2}\left(\Omega\right)}x}\sin\left(\frac{n\pi}{L}x\right)$$

Solving gives the following relation for ω_n:

$$\begin{aligned}
\sqrt{\frac{-\omega_n^2 C^2(\Omega)+\frac{\Delta}{r^2}k_y^2+k_z^2}{\gamma_2}} &= \frac{n\pi}{L} \\
\frac{-\omega_n^2 C^2(\Omega)+\frac{\Delta}{r^2}k_y^2+k_z^2}{-\frac{r^2}{\Delta}C^{-2}(\Omega)} &= \left(\frac{n\pi}{L}\right)^2 \\
\omega_n^2\frac{\Delta}{r^2}C^4(\Omega)-\frac{\Delta}{r^2}C^2(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) &= \left(\frac{n\pi}{L}\right)^2 \\
\omega_n^2 &= \frac{r^2}{\Delta}C^{-4}(\Omega)\left(\frac{n\pi}{L}\right)^2+C^{-2}(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right) \\
&= \frac{r^2}{\Delta}C^{-4}(\Omega)\left[\left(\frac{n\pi}{L}\right)^2+\frac{\Delta}{r^2}C^2(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\right] \\
\omega_n &= \frac{r}{\sqrt{\Delta}}C^{-2}(\Omega)\sqrt{\left(\frac{n\pi}{L}\right)^2+\frac{\Delta}{r^2}C^2(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)}
\end{aligned}$$

The complete solution is

$$\Psi = C_2 e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}e^{-i\frac{\omega_n A(\omega_d-\Omega)}{r^3C^{-2}(\Omega)}x}\sin\left(\frac{n\pi}{L}x\right)$$

In fact, Ψ is the solution for mode n, rather than the full field. We use u_n in place of Ψ to denote this mode solution:

$$u_n = C_2 e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}e^{-i\frac{\omega_n A(\omega_d-\Omega)}{r^3C^{-2}(\Omega)}x}\sin\left(\frac{n\pi}{L}x\right)$$

or

$$u_n = N_n e^{-i\omega_n t}e^{i\vec{k}_\perp\cdot\vec{x}_\perp}e^{-i\beta_n x}\sin\left(\frac{n\pi}{L}x\right)$$

where $C_2, \beta = \frac{\omega_n C^2(\Omega)A}{r^3}(\omega_d-\Omega)$ correspond to N_n, β_n, respectively.

We now use Ψ to denote the full field again. The quantized field $\hat{\Psi}$ is a linear combination of all modes, with a discrete sum in the x direction and continuous integration in the y, z directions:

$$\hat{\Psi} = \sum_n \int \frac{d^2k_\perp}{(2\pi)^2}\left(\hat{a}_{n,\vec{k}_\perp}u_n+\hat{a}^\dagger_{n,\vec{k}_\perp}u_n^*\right)$$

However, with the δ-function normalization used here, the mode functions satisfy $\langle u_n|u_m\rangle = \delta^{(2)}\left(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\right)\delta_{mn}$, so the normalization coefficient N_n already includes a factor of $\frac{1}{2\pi}$. The momentum integral should therefore be $\int d^2k_\perp$, without an additional division by 2π. Thus,

$$\hat{\Psi} = \sum_n \int d^2k_\perp\left(\hat{a}_{n,\vec{k}_\perp}u_n+\hat{a}^\dagger_{n,\vec{k}_\perp}u_n^*\right)$$

IV. The Klein-Gordon Inner Product, Normalization Conditions, and Integration Constants

$$\langle u_n, u_m\rangle = i\int_S\left[\left(\partial_\mu u_n\right)u_m^*-u_n\left(\partial_\mu u_m^*\right)\right]\sqrt{-\hat{g}_S}n^\mu dS$$

1. The hypersurface S ($dt=0$, $t=$ constant): the unit normal vector to S and its dual; the induced metric matrix on S and its determinant. The induced metric is given by

$$h_{\mu\nu} = g_{\mu\nu}-n_\mu n_\nu (n^\mu n_\mu = 1)$$

S is the hypersurface $t=$ constant, whose normal direction is $n_\mu \propto \partial_\mu t = (1,0,0,0)$. Let the future-directed unit normal to S in the dual space (the normal covector, where a covector is a dual vector) be

$$n_\mu = (C,0,0,0),$$

Here, C is a normalization coefficient to be determined. The corresponding normal vector in the tangent space is

$$n^\mu = \hat{g}^{\mu\nu}n_\nu.$$

We have

$$\begin{aligned} n^t &= \hat{g}^{tt} n_t + \hat{g}^{tx} n_x + \hat{g}^{ty} n_y + \hat{g}^{tz} n_z \\ &= \hat{g}^{tt} n_t + 0 \\ &= \frac{A}{\Delta r^2} C, \end{aligned}$$

$$n^x = \hat{g}^{xt} n_t = \frac{A}{\Delta r} \left(\omega_d - \Omega\right) C,$$

$$n^y = \hat{g}^{yt} n_t = 0,$$

$$n^z = \hat{g}^{zt} n_t = 0.$$

A unit vector has norm 1 and therefore satisfies

$$\begin{aligned} n_\mu n^\mu &= 1 \\ n_t n^t + n_x n^x + n_y n^y + n_z n^z &= 1 \\ \frac{A}{\Delta r^2} C^2 &= 1 \\ C &= \sqrt{\frac{\Delta r^2}{A}} \\ &= r\sqrt{\frac{\Delta}{A}}, \end{aligned}$$

This gives

$$n^\mu = r\sqrt{\frac{\Delta}{A}} \left(\frac{A}{\Delta r^2}, \frac{A}{\Delta r} \left(\omega_d - \Omega\right), 0, 0 \right).$$

(Both n_t and n^t are greater than 0, so the normal is future-directed.)

From

$$\begin{aligned} \hat{g}_{S,\mu\nu} &= \hat{g}_{\mu\nu} - n_\mu n_\nu \\ &= \hat{g}_{\mu\nu} - \left(r\sqrt{\frac{\Delta}{A}} \right)^2 \delta_{\mu t} \delta_{\nu t} \end{aligned}$$

we obtain

$$\hat{g}_{S,ij} = \hat{g}_{ij}.$$

Alternatively, starting from

$$ds^2 = \hat{g}_{tt} dt^2 + 2\hat{g}_{tx} dt dx + \hat{g}_{yy} dy^2 + \hat{g}_{zz} dz^2 + \hat{g}_{xx} dx^2,$$

and setting

$$dt = 0,$$

gives

$$ds^{'2} = \hat{g}_{xx} dx^2 + \hat{g}_{yy} dy^2 + \hat{g}_{zz} dz^2.$$

The induced metric on this hypersurface ($t =$ constant) is

$$\begin{aligned} \hat{g}_{S,ij} &= \hat{g}_{ij} \\ &= \begin{pmatrix} \hat{g}_{xx} & 0 & 0 \\ 0 & \hat{g}_{yy} & 0 \\ 0 & 0 & \hat{g}_{zz} \end{pmatrix}, \end{aligned}$$

and its determinant is

$$\begin{aligned} \hat{g}_S = \hat{g}_{xx} \cdot \hat{g}_{yy} \cdot \hat{g}_{zz} &= -\frac{A}{r^4} \cdot \left(-\frac{r^2}{\Delta} \right) \cdot (-1) \\ &= -\frac{A}{\Delta r^2}. \end{aligned}$$

2. The hypersurface $\tilde{S}$ ($d\tilde{t} = 0$, $\tilde{t} =$ constant): the unit normal vector to $\tilde{S}$ and its dual; the induced metric matrix on $\tilde{S}$ and its determinant.

Introduce a coordinate transformation to diagonalize the metric matrix. The four-dimensional spacetime line element is

$$\begin{aligned} ds^2 &= \hat{g}_{tt}dt^2 + 2\hat{g}_{tx}dtdx + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2 + \hat{g}_{xx}dx^2 \\ &= \hat{g}_{tt}\left(dt^2 + 2\frac{\hat{g}_{tx}}{\hat{g}_{tt}}dtdx + \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}^2}dx^2\right) - \hat{g}_{tt}\frac{\hat{g}_{tx}^2}{\hat{g}_{tt}^2}dx^2 + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2 + \hat{g}_{xx}dx^2 \\ &= \hat{g}_{tt}\left(dt + \frac{\hat{g}_{tx}}{\hat{g}_{tt}}dx\right)^2 + \left(\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right)dx^2 + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2, \end{aligned}$$

The coordinate transformation $d\tilde{t} = dt + \frac{\hat{g}_{tx}}{\hat{g}_{tt}}dx$ diagonalizes the metric matrix. Setting

$$d\tilde{t} = 0,$$

gives the induced metric matrix on the hypersurface $\tilde{S}$ ($\tilde{t}$ = constant):

$$\hat{g}_{\tilde{S},ij} = \begin{pmatrix} \hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}} & 0 & 0 \\ 0 & \hat{g}_{yy} & 0 \\ 0 & 0 & \hat{g}_{zz} \end{pmatrix},$$

Its determinant is

$$\begin{aligned} \hat{g}_{\tilde{S}} &= \left(\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right) \times \hat{g}_{yy} \times \hat{g}_{zz} \\ &= \frac{\hat{g}}{\hat{g}_{tt}} = \frac{-1}{C^{-2}(\Omega)} = -C^2(\Omega). \end{aligned}$$

(using $\hat{g} = \det(\hat{g}_{\mu\nu}) = \left(\hat{g}_{tt}\hat{g}_{xx} - \hat{g}_{tx}^2\right) \times \hat{g}_{yy} \times \hat{g}_{zz} = -1$).

The use of $\hat{g}_S = -\frac{\hat{g}}{\hat{g}_{tt}}$ in the paper can be explained as follows. The metric signature is $(+,-,-,-)$. For the spatial part, $(-1)\times(-1)\times(-1) = -1$, so the determinant of the spatial metric, $\hat{g}_S$, should be < 0. Since the quantity under the square root in the spatial volume element $\sqrt{\hat{g}_S}dS$ cannot be negative, a minus sign must be included. Thus, the spatial volume element used here is $\sqrt{-\hat{g}_S}dS$.

Method 2 for finding the induced metric on the hypersurface $\tilde{S}$ ($\tilde{t}$ = constant): the metric

$$\begin{aligned} ds^2 &= C^{-2}(\Omega)\,dt^2 - 2\frac{A}{r^3}(\Omega - \omega_d)\,dtdx - \frac{r^2}{\Delta}dy^2 - dz^2 - \frac{A}{r^4}dx^2 \\ &= \hat{g}_{tt}dt^2 + 2\hat{g}_{tx}dtdx + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2 + \hat{g}_{xx}dx^2 \end{aligned}$$

contains the cross term $dtdx$, so the time and spatial coordinate directions are not orthogonal. Construct a direction orthogonal to the time direction, such that the inner product of a vector in this direction with a vector in the time direction is 0. The basis vector in the time direction is

$$\partial_t = (\partial_t)^\mu \partial_\mu$$

From

$$\partial_t f(x^\mu) = \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x^\mu}\frac{\partial x^\mu}{\partial t}$$

we obtain

$$\partial_t = \frac{\partial x^\mu}{\partial t}\frac{\partial}{\partial x^\mu} = \frac{\partial x^\mu}{\partial t}\partial_\mu$$

and hence

$$(\partial_t)^\mu = \frac{\partial x^\mu}{\partial t} = (1,0,0,0) = \delta_t^\mu$$

For an infinitesimal displacement

$$d\boldsymbol{x} = dx^\nu \partial_\nu$$

require the inner product to vanish:

$$\begin{aligned} \hat{g}_{\mu\nu}(\partial_t)^\mu dx^\nu &= \hat{g}_{\mu\nu}\delta_t^\mu dx^\nu \\ &= \hat{g}_{t\nu}dx^\nu \\ &= \hat{g}_{tt}dt + \hat{g}_{ti}dx^i \\ &= 0 \end{aligned}$$

This gives

$$dt = -\frac{\hat{g}_{ti}}{\hat{g}_{tt}}dx^i$$

The line element on this hypersurface $\tilde{S}$, which is orthogonal to the time direction, is

$$\begin{aligned}
d\tilde{s}^2 &= \hat{g}_{tt}dt^2 + 2\hat{g}_{tx}dtdx + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2 + \hat{g}_{xx}dx^2 \\
&= \hat{g}_{tt}\left(-\frac{\hat{g}_{ti}}{\hat{g}_{tt}}dx^i\right)^2 + 2\hat{g}_{tx}\left(-\frac{\hat{g}_{ti}}{\hat{g}_{tt}}dx^i\right)dx + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2 + \hat{g}_{xx}dx^2 \\
&= \hat{g}_{tt}\left(-\frac{\hat{g}_{tx}}{\hat{g}_{tt}}dx\right)^2 + 2\hat{g}_{tx}\left(-\frac{\hat{g}_{tx}}{\hat{g}_{tt}}dx\right)dx + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2 + \hat{g}_{xx}dx^2 \\
&= \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}dx^2 - 2\frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}dx^2 + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2 + \hat{g}_{xx}dx^2 \\
&= \left(\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right)dx^2 + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2
\end{aligned}$$

We now find the unit normal vector to $\tilde{S}$ and its dual. The metric components are treated as constants, so $d\tilde{t} = dt + \frac{\hat{g}_{tx}}{\hat{g}_{tt}}dx$ is integrable, giving $\tilde{t} = t + \frac{\hat{g}_{tx}}{\hat{g}_{tt}}x + \text{const}$. The hypersurface $\tilde{S}$ is defined by $\tilde{t} = \text{constant}$, and its normal direction is $n_\mu \propto \partial_\mu \tilde{t} = \left(1, \frac{\hat{g}_{tx}}{\hat{g}_{tt}}, 0, 0\right)$. Let the future-directed unit normal to $\tilde{S}$ in the dual space (the normal covector, where a covector is a dual vector; see Liang Canbin, p. 99) be

$$n_\mu = \tilde{C}\left(1, \frac{\hat{g}_{tx}}{\hat{g}_{tt}}, 0, 0\right).$$

Here, $\tilde{C}$ is a normalization coefficient to be determined.

$$n^\mu = \hat{g}^{\mu\nu}n_\nu,$$

$$\hat{g}_{\mu\nu} = \begin{pmatrix} C^{-2}(\Omega) & -\frac{A}{r^3}(\Omega - \omega_d) & 0 & 0 \\ -\frac{A}{r^3}(\Omega - \omega_d) & -\frac{A}{r^4} & 0 & 0 \\ 0 & 0 & -\frac{r^2}{\Delta} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

$$\hat{g}^{\mu\nu} = \begin{pmatrix} \frac{A}{\Delta r^2} & \frac{A}{\Delta r}(\omega_d - \Omega) & 0 & 0 \\ \frac{A}{\Delta r}(\omega_d - \Omega) & -\frac{r^2}{\Delta}C^{-2}(\Omega) & 0 & 0 \\ 0 & 0 & -\frac{\Delta}{r^2} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

$$C(\Omega) = \left\{\frac{\Delta r^2}{A}\left[1 - \frac{A^2}{\Delta r^4}(\Omega - \omega_d)^2\right]\right\}^{-1/2},$$

We have

$$\begin{aligned}
n^t &= \hat{g}^{tt}n_t + \hat{g}^{tx}n_x + \hat{g}^{ty}n_y + \hat{g}^{tz}n_z \\
&= \frac{A}{\Delta r^2}\cdot\tilde{C} + \frac{A}{\Delta r}(\omega_d - \Omega)\cdot\frac{-\frac{A}{r^3}(\Omega - \omega_d)}{C^{-2}(\Omega)}\tilde{C} + 0 + 0 \\
&= \frac{A}{\Delta r^2}\tilde{C} + \frac{A^2}{\Delta r^4}(\Omega - \omega_d)^2\frac{1}{\frac{\Delta r^2}{A}\left[1 - \frac{A^2}{\Delta r^4}(\Omega - \omega_d)^2\right]}\tilde{C} \\
&= \frac{A}{\Delta r^2}\tilde{C} + \frac{A}{\Delta r^2}\left[-1 + \frac{1}{1 - \frac{A^2}{\Delta r^4}(\Omega - \omega_d)^2}\right]\tilde{C} \\
&= \frac{A}{\Delta r^2}\frac{1}{1 - \frac{A^2}{\Delta r^4}(\Omega - \omega_d)^2}\tilde{C},
\end{aligned}$$

where

$$\frac{a}{1-a} = -1 + \frac{1}{1-a}, a = \frac{A^2}{\Delta r^4}(\Omega - \omega_d)^2.$$

$$\begin{aligned}
n^x &= \hat{g}^{xt}n_t + \hat{g}^{xx}n_x + \hat{g}^{xy}n_y + \hat{g}^{xz}n_z \\
&= \frac{A}{\Delta r}(\omega_d - \Omega)\cdot\tilde{C} - \frac{r^2}{\Delta}C^{-2}(\Omega)\cdot\frac{-\frac{A}{r^3}(\Omega - \omega_d)}{C^{-2}(\Omega)}\tilde{C} + 0 + 0 \\
&= \frac{A}{\Delta r}(\omega_d - \Omega)\tilde{C} + \frac{r^2}{\Delta}\frac{A}{r^3}(\Omega - \omega_d)\tilde{C} \\
&= -\frac{A}{\Delta r}(\Omega - \omega_d)\tilde{C} + \frac{A}{\Delta r}(\Omega - \omega_d)\tilde{C} \\
&= 0,
\end{aligned}$$

$$n^y = n^z = 0, \left(\hat{g}^{yt} = \hat{g}^{yt} = \hat{g}^{zt} = \hat{g}^{zt} = 0, n_y = n_z = 0\right).$$

A unit vector has norm 1 and therefore satisfies

$$\begin{aligned} n_\mu n^\mu &= 1 \\ n_t n^t + n_x n^x + n_y n^y + n_z n^z &= 1 \\ \frac{A}{\Delta r^2} \frac{1}{1 - \frac{A^2}{\Delta r^4} \left(\Omega - \omega_d\right)^2} \tilde{C}^2 &= 1 \\ \tilde{C} &= \sqrt{\frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4} \left(\Omega - \omega_d\right)^2\right]} = C^{-1}\left(\Omega\right), \end{aligned}$$

This gives

$$\begin{aligned} n^\mu &= \left(\frac{A}{\Delta r^2} \frac{1}{1 - \frac{A^2}{\Delta r^4} \left(\Omega - \omega_d\right)^2} \cdot \sqrt{\frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4} \left(\Omega - \omega_d\right)^2\right]}, 0, 0, 0\right) \\ &= \left(\frac{1}{\left(\sqrt{\frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4} \left(\Omega - \omega_d\right)^2\right]}\right)^2} \cdot \sqrt{\frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4} \left(\Omega - \omega_d\right)^2\right]}, 0, 0, 0\right) \\ &= \left(\left\{\frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4} \left(\Omega - \omega_d\right)^2\right]\right\}^{-1/2}, 0, 0, 0\right) \\ &= C\left(\Omega\right)\left(1, 0, 0, 0\right). \end{aligned}$$

(Both n_t and n^t are greater than 0, so the normal is future-directed.)

Summary:

1. The hypersurface S ($dt = 0$, $t =$ constant). The unit normal vector to S and its dual are

$$\begin{aligned} n^\mu &= r\sqrt{\frac{\Delta}{A}} \left(\frac{A}{\Delta r^2}, \frac{A}{\Delta r}\left(\omega_d - \Omega\right), 0, 0\right), \\ n_\mu &= \left(r\sqrt{\frac{\Delta}{A}}, 0, 0, 0\right). \end{aligned}$$

The induced metric matrix on S and its determinant are

$$\begin{aligned} \hat{g}_{S,ij} &= \hat{g}_{ij} \\ &= \begin{pmatrix} \hat{g}_{xx} & 0 & 0 \\ 0 & \hat{g}_{yy} & 0 \\ 0 & 0 & \hat{g}_{zz} \end{pmatrix}, \\ \hat{g}_S &= -\frac{A}{\Delta r^2}. \end{aligned}$$

2. The hypersurface $\tilde{S}$ ($d\tilde{t} = 0$, $\tilde{t} =$ constant). The unit normal vector to $\tilde{S}$ and its dual are

$$\begin{aligned} n^\mu &= C\left(\Omega\right)\left(1, 0, 0, 0\right), \\ n_\mu &= C^{-1}\left(\Omega\right)\left(1, \frac{\hat{g}_{tx}}{\hat{g}_{tt}}, 0, 0\right). \end{aligned}$$

The induced metric matrix on $\tilde{S}$ and its determinant are

$$\begin{aligned} \hat{g}_{\tilde{S},ij} &= \begin{pmatrix} \hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}} & 0 & 0 \\ 0 & \hat{g}_{yy} & 0 \\ 0 & 0 & \hat{g}_{zz} \end{pmatrix}, \\ \hat{g}_{\tilde{S}} &= -C^2\left(\Omega\right). \end{aligned}$$

where $C\left(\Omega\right) = \left\{\frac{\Delta r^2}{A}\left[1 - \frac{A^2}{\Delta r^4}\left(\Omega - \omega_d\right)^2\right]\right\}^{-1/2}$, $\hat{g}_{tx} = -\frac{A}{r^3}\left(\Omega - \omega_d\right)$, $\hat{g}_{tt} = C^{-2}\left(\Omega\right)$.

The quantum field modes satisfy the normalization condition

$$\langle u_n, u_m \rangle = i \int_S \left[\left(\partial_\mu u_n\right) u_m^* - u_n \left(\partial_\mu u_m^*\right)\right] \sqrt{-\hat{g}_S} n^\mu dS = \delta^{(2)}\left(\vec{k}_{\perp,n} - \vec{k}_{\perp,m}\right) \delta_{mn}.$$

1. The hypersurface S ($dt = 0$, $t =$ constant):

$$\begin{aligned}
\langle u_n, u_m\rangle &= i\int_S\Big[\Big(\partial_\mu u_n\Big)u_m^* - u_n\Big(\partial_\mu u_m^*\Big)\Big]\sqrt{-\hat{g}_S}n^\mu dS \\
&= i\int_S\Big[\Big(\partial_t u_n\Big)u_m^* - u_n\Big(\partial_t u_m^*\Big)\Big]\sqrt{-\hat{g}_S}n^t dS + i\int_S\Big[\Big(\partial_x u_n\Big)u_m^* - u_n\Big(\partial_x u_m^*\Big)\Big]\sqrt{-\hat{g}_S}n^x dS \\
&= i\int_S\Big[\Big(-i\omega_n u_n\Big)u_m^* - u_n\Big(i\omega_m u_m^*\Big)\Big]\sqrt{-\hat{g}_S}n^t dS + i\int_S\Big[\Big(\partial_x u_n\Big)u_m^* - u_n\Big(\partial_x u_m^*\Big)\Big]\sqrt{-\hat{g}_S}n^x dS \\
&= i\int_S\Big[-i\Big(\omega_n+\omega_m\Big)u_n u_m^*\Big]\sqrt{-\hat{g}_S}n^t dS + i\int_S\Big[\Big(-i\beta_n u_n + \frac{n\pi}{L}\cos\Big(\frac{n\pi}{L}x\Big)F_n\Big)u_m^* \\
&\quad - u_n\Big(i\beta_m u_m^* + \frac{m\pi}{L}\cos\Big(\frac{m\pi}{L}x\Big)F_m^*\Big)\Big]\sqrt{-\hat{g}_S}n^x dS \\
&= i\int_S\Big[-i\Big(\omega_n+\omega_m\Big)u_n u_m^*\Big]\sqrt{-\hat{g}_S}n^t dS + i\int_S\Big[\Big(-i\beta_n u_n u_m^* + \frac{n\pi}{L}\cos\Big(\frac{n\pi}{L}x\Big)F_n u_m^* \\
&\quad \Big) - \Big(i\beta_m u_n u_m^* + \frac{m\pi}{L}\cos\Big(\frac{m\pi}{L}x\Big)F_m^* u_n\Big)\Big]\sqrt{-\hat{g}_S}n^x dS \\
&= i\int_S\Big[-i\Big(\omega_n+\omega_m\Big)u_n u_m^*\Big]\sqrt{-\hat{g}_S}n^t dS + i\int_S\Big[-i\Big(\beta_n+\beta_m\Big)u_n u_m^*\Big]\sqrt{-\hat{g}_S}n^x dS \\
&\quad + i\int_S\Big[\frac{n\pi}{L}\cos\Big(\frac{n\pi}{L}x\Big)F_n u_m^* - \frac{m\pi}{L}\cos\Big(\frac{m\pi}{L}x\Big)F_m^* u_n\Big]\sqrt{-\hat{g}_S}n^x dS
\end{aligned}$$

where

$$\begin{aligned}
\partial_x u_n &= \partial_x\Big[N_n e^{-i\omega_n t}e^{i\vec{k}_{\perp,n}\cdot\vec{x}_\perp}e^{-i\beta_n x}\sin\Big(\frac{n\pi}{L}x\Big)\Big] \\
&= -i\beta_n u_n + \frac{n\pi}{L}\cos\Big(\frac{n\pi}{L}x\Big)N_n e^{-i\omega_n t}e^{i\vec{k}_{\perp,n}\cdot\vec{x}_\perp}e^{-i\beta_n x} \\
&= -i\beta_n u_n + \frac{n\pi}{L}\cos\Big(\frac{n\pi}{L}x\Big)F_n
\end{aligned}$$

$$\begin{aligned}
\partial_x u_m^* &= \partial_x\Big[N_m e^{i\omega_m t}e^{-i\vec{k}_{\perp,m}\cdot\vec{x}_\perp}e^{i\beta_m x}\sin\Big(\frac{m\pi}{L}x\Big)\Big] \\
&= i\beta_m u_m^* + \frac{m\pi}{L}\cos\Big(\frac{m\pi}{L}x\Big)F_m^*
\end{aligned}$$

and

$$\begin{aligned}
&i\int_S\Big[-i\Big(\omega_n+\omega_m\Big)u_n u_m^*\Big]\sqrt{-\hat{g}_S}n^t dS = i\int_S\Big\{-i\Big(\omega_n+\omega_m\Big)\Big[N_n e^{-i\omega_n t}e^{i\vec{k}_{\perp,n}\cdot\vec{x}_\perp} \\
&\quad e^{-i\beta_n x}\sin\Big(\frac{n\pi}{L}x\Big)\Big]\Big[N_m e^{i\omega_m t}e^{-i\vec{k}_{\perp,m}\cdot\vec{x}_\perp}e^{i\beta_m x}\sin\Big(\frac{m\pi}{L}x\Big)\Big]\Big\}\sqrt{-\hat{g}_S}n^t dxdydz \\
&= i\int_S\Big\{-i\Big(\omega_n+\omega_m\Big)\Big[N_n N_m e^{-i\Big(\omega_n-\omega_m\Big)t}e^{i\Big(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\Big)\cdot\vec{x}_\perp}e^{-i\Big(\beta_n-\beta_m\Big)x}\sin\Big(\frac{n\pi}{L}x\Big) \\
&\quad \sin\Big(\frac{m\pi}{L}x\Big)\Big]\Big\}\sqrt{-\hat{g}_S}n^t dxdydz \\
&= \Big(\omega_n+\omega_m\Big)\sqrt{-\hat{g}_S}n^t N_n N_m e^{-i\Big(\omega_n-\omega_m\Big)t}\int_0^L e^{-i\Big(\beta_n-\beta_m\Big)x}\sin\Big(\frac{n\pi}{L}x\Big)\sin\Big(\frac{m\pi}{L}x\Big) \\
&\quad dx\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}e^{i\Big(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\Big)\cdot\vec{x}_\perp}dydz
\end{aligned}$$

Similarly,

$$\begin{aligned}
&i\int_S\Big[-i\Big(\beta_n+\beta_m\Big)u_n u_m^*\Big]\sqrt{-\hat{g}_S}n^x dS = \Big(\beta_n+\beta_m\Big)\sqrt{-\hat{g}_S}n^x N_n N_m e^{-i\Big(\omega_n-\omega_m\Big)t} \\
&\quad \int_0^L e^{-i\Big(\beta_n-\beta_m\Big)x}\sin\Big(\frac{n\pi}{L}x\Big)\sin\Big(\frac{m\pi}{L}x\Big)dx\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}e^{i\Big(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\Big)\cdot\vec{x}_\perp}dydz
\end{aligned}$$

Also,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{i\left(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\right)\cdot\vec{x}_{\perp}}dydz = (2\pi)^2\,\delta^{(2)}\left(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\right)$$

Imposing $\langle u_n, u_m\rangle = \delta^{(2)}\left(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\right)\delta_{mn}$ gives

$$2\omega_n\sqrt{-\hat{g}_S}n^t N_n^2\frac{L}{2}(2\pi)^2 + 2\beta_n\sqrt{-\hat{g}_S}n^x N_n^2\frac{L}{2}(2\pi)^2 = 1$$

(For $n=m$, and only in this case, $\delta_{mn}=1$; when $n\neq m$, we have $\delta_{mn}=0$ and correspondingly $\langle u_n,u_m\rangle=0$. Also, $\omega_n+\omega_m\to 2\omega_n$; $e^{-i(\omega_n-\omega_m)t}\to e^{-i\cdot 0\cdot t}=1$; $\int_0^L e^{-i(\beta_n-\beta_m)x}\sin\left(\frac{n\pi}{L}x\right)\sin\left(\frac{m\pi}{L}x\right)dx \to \int_0^L \sin^2\left(\frac{n\pi}{L}x\right)dx = L/2$.) We obtain

$$\begin{aligned}
N_n^2 &= \frac{1}{2\omega_n\sqrt{-\hat{g}_S}n^t\frac{L}{2}(2\pi)^2 + 2\beta_n\sqrt{-\hat{g}_S}n^x\frac{L}{2}(2\pi)^2}\\
&= \frac{1}{(2\pi)^2 L\sqrt{-\hat{g}_S}\left(\omega_n n^t+\beta_n n^x\right)}\\
&= \frac{1}{(2\pi)^2 L\sqrt{\frac{A}{\Delta r^2}}\left\{\omega_n r\sqrt{\frac{\Delta}{A}}\frac{A}{\Delta r^2}+\left[\frac{\omega_n C^2(\Omega)A}{r^3}\left(\omega_d-\Omega\right)\right]r\sqrt{\frac{\Delta}{A}}\frac{A}{\Delta r}\left(\omega_d-\Omega\right)\right\}}\\
&= \frac{1}{(2\pi)^2 L\sqrt{\frac{A}{\Delta r^2}}\omega_n\left\{\sqrt{\frac{A}{\Delta}}\frac{1}{r}+\frac{C^2(\Omega)A}{r^3}\sqrt{\frac{A}{\Delta}}\left(\omega_d-\Omega\right)^2\right\}}\\
&= \frac{1}{(2\pi)^2 L\sqrt{\frac{A}{\Delta r^2}}\omega_n\left\{\sqrt{\frac{A}{\Delta}}\frac{1}{r}+\frac{\left\{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]\right\}^{-1}A}{r^3}\sqrt{\frac{A}{\Delta}}\left(\omega_d-\Omega\right)^2\right\}}\\
&= \frac{1}{(2\pi)^2 L\sqrt{\frac{A}{\Delta r^2}}\omega_n\frac{1}{r}\sqrt{\frac{A}{\Delta}}\left\{1+\frac{1}{\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]}\frac{A^2}{\Delta r^4}\left(\omega_d-\Omega\right)^2\right\}}\\
&= \frac{1}{(2\pi)^2 L\sqrt{\frac{A}{\Delta r^2}}\omega_n\frac{1}{r}\sqrt{\frac{A}{\Delta}}\left\{1+\frac{1}{\frac{A}{\Delta r^2}C^{-2}(\Omega)}\left[1-\frac{A}{\Delta r^2}C^{-2}(\Omega)\right]\right\}}\\
&= \frac{1}{(2\pi)^2 L\sqrt{\frac{A}{\Delta r^2}}\omega_n\frac{1}{r}\sqrt{\frac{A}{\Delta}}\left\{1+\frac{\Delta r^2}{A}C^2(\Omega)\left[1-\frac{A}{\Delta r^2}C^{-2}(\Omega)\right]\right\}}\\
&= \frac{1}{(2\pi)^2 L\sqrt{\frac{A}{\Delta r^2}}\omega_n\frac{1}{r}\sqrt{\frac{A}{\Delta}}\left[\frac{\Delta r^2}{A}C^2(\Omega)\right]}\\
&= \frac{1}{(2\pi)^2 L\frac{A}{\Delta}\frac{1}{r}\omega_n\frac{1}{r}\left[\frac{\Delta r^2}{A}C^2(\Omega)\right]}\\
&= \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)}
\end{aligned}$$

or equivalently,

$$N_n^2 = \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)}$$

Here,

$$C(\Omega) = \left\{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2\right]\right\}^{-1/2}$$

$$C^{-2}(\Omega) = \frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2\right]$$

which give

$$1-\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2 = \frac{A}{\Delta r^2}C^{-2}(\Omega)$$

$$\frac{A^2}{\Delta r^4}\left(\Omega-\omega_d\right)^2 = 1-\frac{A}{\Delta r^2}C^{-2}(\Omega)$$

We also have

$$\beta_n = \frac{b_n A}{r^3}\left(\omega_d-\Omega\right) = \frac{\omega_n C^2(\Omega)A}{r^3}\left(\omega_d-\Omega\right)$$

and

$$\begin{aligned}\int_0^L e^{-i(\beta_n-\beta_n)x}\sin\left(\frac{n\pi}{L}x\right)\sin\left(\frac{n\pi}{L}x\right)dx &= \int_0^L \sin^2\left(\frac{n\pi}{L}x\right)dx \\ &= \int_0^L \frac{1-\cos\left(2\frac{n\pi}{L}x\right)}{2}dx \\ &= \frac{L}{2}, \text{where } \sin 2n\pi = 0\end{aligned}$$

$$F_n u_n^* = F_n^* u_n$$
$$\frac{n\pi}{L}\cos\left(\frac{n\pi}{L}x\right)F_n u_n^* - \frac{n\pi}{L}\cos\left(\frac{n\pi}{L}x\right)F_n^* u_n = 0$$

Note that the original paper gives

$$\begin{aligned}N_n^2 &= \frac{1}{(2\pi)^2 L\omega_n}\frac{1}{r}\sqrt{\frac{A}{\Delta}}C^{-3}(\Omega) \\ &= \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)}\frac{1}{r}\sqrt{\frac{A}{\Delta}}C^{-1}(\Omega) \\ &= \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)}\frac{1}{r}\sqrt{\frac{A}{\Delta}}\left\{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]\right\}^{1/2} \\ &= \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)}\frac{1}{r}\sqrt{\frac{A}{\Delta}}\sqrt{\frac{\Delta}{A}}r\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]^{1/2} \\ &= \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)}\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]^{1/2}\end{aligned}$$

where

$$C(\Omega) = \left\{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]\right\}^{-1/2}$$

2. The hypersurface $\tilde{S}$ ($d\tilde{t}=0$, $\tilde{t}=$ constant):

$$n^\mu = C(\Omega)(1,0,0,0),$$

$$\begin{aligned}\langle u_n, u_m\rangle &= i\int_{\tilde{S}}\left[(\partial_\mu u_n)u_m^* - u_n(\partial_\mu u_m^*)\right]\sqrt{-\hat{g}_{\tilde{S}}}n^\mu d\tilde{S} \\ &= i\int_{\tilde{S}}\left[(\partial_t u_n)u_m^* - u_n(\partial_t u_m^*)\right]\sqrt{-\hat{g}_{\tilde{S}}}n^t d\tilde{S} \\ &= i\int_{\tilde{S}}\left[(-i\omega_n u_n)u_m^* - u_n(i\omega_m u_m^*)\right]\sqrt{-\hat{g}_{\tilde{S}}}n^t d\tilde{S} \\ &= i\int_{\tilde{S}}\left[-i(\omega_n+\omega_m)u_n u_m^*\right]\sqrt{-\hat{g}_{\tilde{S}}}n^t d\tilde{S}\end{aligned}$$

and

$$\begin{aligned}&i\int_{\tilde{S}}\left[-i\Big(\omega_n+\omega_m\Big)u_n u_m^*\right]\sqrt{-\hat{g}_{\tilde{S}}}n^t d\tilde{S} = i\int_{\tilde{S}}\left\{-i\Big(\omega_n+\omega_m\Big)\left[N_n e^{-i\omega_n t}e^{i\vec{k}_{\perp,n}\cdot\vec{x}_\perp}\right.\right. \\ &\quad \left.\left. e^{-i\beta_n x}\sin\Big(\frac{n\pi}{L}x\Big)\right]\left[N_m e^{i\omega_m t}e^{-i\vec{k}_{\perp,m}\cdot\vec{x}_\perp}e^{i\beta_m x}\sin\Big(\frac{m\pi}{L}x\Big)\right]\right\}\sqrt{-\hat{g}_{\tilde{S}}}n^t dxdydz \\ &= i\int_{\tilde{S}}\left\{-i\Big(\omega_n+\omega_m\Big)\left[N_n N_m e^{-i\Big(\omega_n-\omega_m\Big)t}\underset{e}{}^{i\Big(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\Big)\cdot\vec{x}_\perp}\underset{e}{}^{-i\Big(\beta_n-\beta_m\Big)x}\sin\Big(\frac{n\pi}{L}x\Big)\right.\right. \\ &\quad \left.\left.\sin\Big(\frac{m\pi}{L}x\Big)\right]\right\}\sqrt{-\hat{g}_{\tilde{S}}}n^t dxdydz \\ &= \Big(\omega_n+\omega_m\Big)\sqrt{-\hat{g}_{\tilde{S}}}n^t N_n N_m\int_0^L \underset{e}{}^{-i\Big(\omega_n-\omega_m\Big)t(x)}\underset{e}{}^{-i\Big(\beta_n-\beta_m\Big)x}\sin\Big(\frac{n\pi}{L}x\Big)\sin\Big(\frac{m\pi}{L}x\Big) \\ &\quad dx\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\underset{e}{}^{i\Big(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\Big)\cdot\vec{x}_\perp}dydz\end{aligned}$$

where $t(x) = \tilde{t} - \frac{\hat{g}_{tx}}{\hat{g}_{tt}} x - \text{const}$. Also,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{i\left(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\right)\cdot\vec{x}_{\perp}} dydz = (2\pi)^2 \delta^{(2)}\left(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\right)$$

Imposing $\langle u_n, u_m\rangle = \delta^{(2)}\left(\vec{k}_{\perp,n}-\vec{k}_{\perp,m}\right)\delta_{mn}$ gives

$$2\omega_n\sqrt{-\hat{g}_{\tilde{S}}}n^t N_n^2 \frac{L}{2}(2\pi)^2 = 1$$

(For $n = m$, and only in this case, $\delta_{mn} = 1$; when $n \neq m$, we have $\delta_{mn} = 0$ and correspondingly $\langle u_n, u_m\rangle = 0$. Also, $\omega_n + \omega_m \to 2\omega_n$; $e^{-i(\omega_n-\omega_m)t(x)} \to e^{-i\cdot 0\cdot t(x)} = 1$; $\int_0^L e^{-i(\beta_n-\beta_m)x}\sin\left(\frac{n\pi}{L}x\right)\sin\left(\frac{m\pi}{L}x\right)dx \to \int_0^L \sin^2\left(\frac{n\pi}{L}x\right)dx = L/2$.) This gives

$$\begin{aligned} N_n^2 &= \frac{1}{2\omega_n\sqrt{-\hat{g}_{\tilde{S}}}n^t\frac{L}{2}(2\pi)^2} \\ &= \frac{1}{(2\pi)^2 L\omega_n\sqrt{-\hat{g}_{\tilde{S}}}n^t} \\ &= \frac{1}{(2\pi)^2 L\omega_n C(\Omega)\cdot C(\Omega)} \\ &= \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)} \end{aligned}$$

V. Mean Vacuum Energy Density

$$\begin{aligned} \langle\epsilon_{\text{vac}}\rangle &= \frac{1}{V_p}\int_V \sqrt{-\hat{g}_S}dV\langle 0|T_{\mu\nu}\hat{w}^\mu\hat{w}^\nu|0\rangle \\ &= C^2(\Omega)\frac{1}{V_p}\int_V \sqrt{-\hat{g}_S}dV\langle 0|T_{tt}|0\rangle \\ &= C^2(\Omega)\frac{1}{\sqrt{-\hat{g}_S}\int_0^L dx\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz}\sqrt{-\hat{g}_S}\int_V dV\langle 0|T_{tt}|0\rangle \\ &= C^2(\Omega)\frac{1}{L\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz}\int_V dV\langle 0|T_{tt}|0\rangle \\ &= C^2(\Omega)\frac{1}{L\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz}\int_0^L dx\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz\langle 0|T_{tt}|0\rangle \end{aligned}$$

where

$$T_{\mu\nu}\hat{w}^\mu\hat{w}^\nu = T_{tt}\hat{w}^t\hat{w}^t = C^2(\Omega)T_{tt}$$

$$\hat{w}^\mu = C(\Omega)(1,0,0,0)$$

$$V_p = \int_V \sqrt{-\hat{g}_S}dV = \sqrt{-\hat{g}_S}\int_0^L dx\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz$$

We have

$$\begin{aligned} T_{\mu\nu} &= 2\frac{\partial\mathcal{L}}{\partial g^{\mu\nu}} - g_{\mu\nu}\mathcal{L} \\ &= 2\frac{\partial\left(\frac{1}{2}g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi\right)}{\partial g^{\mu\nu}} - g_{\mu\nu}\left(\frac{1}{2}g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi\right) \\ &= \frac{\partial g^{\alpha\beta}}{\partial g^{\mu\nu}}\partial_\alpha\Psi\partial_\beta\Psi - \frac{1}{2}g_{\mu\nu}\left(g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi\right) \\ &= \frac{1}{2}\left(\delta_\mu^\alpha\delta_\nu^\beta + \delta_\nu^\alpha\delta_\mu^\beta\right)\partial_\alpha\Psi\partial_\beta\Psi - \frac{1}{2}g_{\mu\nu}\left(g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi\right) \\ &= \frac{1}{2}\left(\partial_\mu\Psi\partial_\nu\Psi + \partial_\nu\Psi\partial_\mu\Psi\right) - \frac{1}{2}g_{\mu\nu}\left(g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi\right) \\ &= \partial_\mu\Psi\partial_\nu\Psi - \frac{1}{2}g_{\mu\nu}\left(g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi\right) \end{aligned}$$

where, for a massless scalar field in vacuum in curved spacetime, the Lagrangian density is

$$\mathcal{L} = \frac{1}{2} g^{\alpha\beta} \partial_\alpha \Psi \partial_\beta \Psi$$

(With the metric signature $(-,+,+,+)$, the kinetic term in the Lagrangian density must be defined as

$$-\frac{1}{2} g^{\mu\nu} \partial_\mu \Psi \partial_\nu \Psi$$

to obtain positive kinetic energy. To ensure that the energy density measured by a local observer is positive, that is, $T_{\mu\nu} U^\mu U^\nu > 0$, the energy-momentum tensor must be defined as $T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}$, where $S_m = \int d^4x \sqrt{-g} \mathcal{L}_m$ is the matter action. This gives $T_{\mu\nu} = -2 \frac{\partial \mathcal{L}_m}{\partial g^{\mu\nu}} + g_{\mu\nu} \mathcal{L}_m$. With the metric signature $(+,-,-,-)$, the kinetic term in the Lagrangian density must instead be defined as $\frac{1}{2} g^{\mu\nu} \partial_\mu \Psi \partial_\nu \Psi$ to obtain positive kinetic energy. To ensure that the energy density measured by a local observer is positive, that is, $T_{\mu\nu} U^\mu U^\nu > 0$, the energy-momentum tensor must be defined as $T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}$, giving $T_{\mu\nu} = 2 \frac{\partial \mathcal{L}_m}{\partial g^{\mu\nu}} - g_{\mu\nu} \mathcal{L}_m$.) We also use

$$\frac{\partial g^{\alpha\beta}}{\partial g^{\mu\nu}} = \frac{1}{2} \left(\delta^\alpha_\mu \delta^\beta_\nu + \delta^\alpha_\nu \delta^\beta_\mu \right)$$

Then

$$\begin{aligned}
\hat{T}_{tt} &= \partial_t \hat{\Psi} \partial_t \hat{\Psi} - \frac{1}{2} \hat{g}_{tt} \left(\hat{g}^{\alpha\beta} \partial_\alpha \hat{\Psi} \partial_\beta \hat{\Psi} \right) \\
&= \partial_t \sum_n \int d^2 k_{\perp,n} \left(\hat{a}_{n,\boldsymbol{k}_{\perp,n}} u_n + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} u^*_n \right) \partial_t \sum_m \int d^2 k_{\perp,m} \left(\hat{a}_{m,\boldsymbol{k}_{\perp,m}} u_m + \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} u^*_m \right) \\
&\quad - \frac{1}{2} \hat{g}_{tt} \left[\hat{g}^{\mu\nu} \partial_\mu \sum_n \int d^2 k_{\perp,n} \left(\hat{a}_{n,\boldsymbol{k}_{\perp,n}} u_n + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} u^*_n \right) \partial_\nu \sum_m \int d^2 k_{\perp,m} \left(\hat{a}_{m,\boldsymbol{k}_{\perp,m}} u_m \right. \right. \\
&\quad \left. \left. + \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} u^*_m \right) \right] \\
&= \sum_n \int d^2 k_{\perp,n} \left(\hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_t u_n + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_t u^*_n \right) \sum_m \int d^2 k_{\perp,m} \left(\hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_t u_m + \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_t \right. \\
&\quad \left. u^*_m \right) - \frac{1}{2} \hat{g}_{tt} \left[\hat{g}^{\mu\nu} \sum_n \int d^2 k_{\perp,n} \left(\hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_\mu u_n + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_\mu u^*_n \right) \sum_m \int d^2 k_{\perp,m} \left(\hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_\nu \right. \right. \\
&\quad \left. \left. u_m + \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_\nu u^*_m \right) \right] \\
&= \sum_{n,m} \int d^2 k_{\perp,n} \int d^2 k_{\perp,m} \left(\hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_t u_n \cdot \hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_t u_m + \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_t u_n \cdot \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_t u^*_m \right. \\
&\quad + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_t u^*_n \cdot \hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_t u_m + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_t u^*_n \cdot \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_t u^*_m \Big) - \frac{1}{2} \hat{g}_{tt} \Big[\hat{g}^{\mu\nu} \sum_{n,m} \int d^2 k_{\perp,n} \\
&\quad \int d^2 k_{\perp,m} \left(\hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_\mu u_n \cdot \hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_\nu u_m + \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_\mu u_n \cdot \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_\nu u^*_m + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_\mu u^*_n \right. \\
&\quad \left. \left. \cdot \hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_\nu u_m + \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_\mu u^*_n \cdot \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_\nu u^*_m \right) \Big]
\end{aligned}$$

where

$$\hat{\Psi} = \sum_n \int d^2 k_\perp \left(\hat{a}_{n,\boldsymbol{k}_\perp} u_n + \hat{a}^\dagger_{n,\boldsymbol{k}_\perp} u^*_n \right)$$

$$\begin{aligned}
\langle 0 | \hat{T}_{tt} | 0 \rangle &= \sum_{n,m} \int d^2 k_{\perp,n} \int d^2 k_{\perp,m} \partial_t u_n \partial_t u^*_m \langle 0 | \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle - \frac{1}{2} \hat{g}_{tt} \Big[\hat{g}^{\mu\nu} \sum_{n,m} \int d^2 \\
&\quad k_{\perp,n} \int d^2 k_{\perp,m} \partial_\mu u_n \partial_\nu u^*_m \langle 0 | \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle \Big] \\
&= \sum_{n,m} \int d^2 k_{\perp,n} \int d^2 k_{\perp,m} \partial_t u_n \partial_t u^*_m \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm} - \frac{1}{2} \hat{g}_{tt} \Big[\hat{g}^{\mu\nu} \sum_{n,m} \int d^2 k_{\perp,n} \\
&\quad \int d^2 k_{\perp,m} \partial_\mu u_n \partial_\nu u^*_m \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm} \Big] \\
&= \sum_n \int d^2 k_{\perp,n} \partial_t u_n \partial_t u^*_n - \frac{1}{2} \hat{g}_{tt} \left(\hat{g}^{\mu\nu} \sum_n \int d^2 k_{\perp,n} \partial_\mu u_n \partial_\nu u^*_n \right)
\end{aligned}$$

$$= \sum_n \int d^2k_{\perp,n} \left[\partial_t u_n \partial_t u_n^* - \frac{1}{2} \hat{g}_{tt} \left(\hat{g}^{\mu\nu} \partial_\mu u_n \partial_\nu u_n^* \right) \right]$$

Here,

$$\langle 0 | \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_t u_n \cdot \hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_t u_m | 0 \rangle = \partial_t u_n \partial_t u_m \langle 0 | \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \hat{a}_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle = 0$$

$$\begin{aligned}
\langle 0 | \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \partial_t u_n \cdot \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_t u_m^* | 0 \rangle &= \partial_t u_n \partial_t u_m^* \langle 0 | \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle \\
&= \partial_t u_n \partial_t u_m^* \langle 0 | \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm} + \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \hat{a}_{n,\boldsymbol{k}_{\perp,n}} | 0 \rangle \\
&= \partial_t u_n \partial_t u_m^* \langle 0 | \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm} | 0 \rangle + \langle 0 | \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \hat{a}_{n,\boldsymbol{k}_{\perp,n}} | 0 \rangle \\
&= \partial_t u_n \partial_t u_m^* \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm} \langle 0 | 0 \rangle + 0 \\
&= \partial_t u_n \partial_t u_m^* \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm}
\end{aligned}$$

$$\langle 0 | \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_t u_n^* \cdot \hat{a}_{m,\boldsymbol{k}_{\perp,m}} \partial_t u_m | 0 \rangle = \partial_t u_n^* \partial_t u_m \langle 0 | \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \hat{a}_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle = 0$$

$$\langle 0 | \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \partial_t u_n^* \cdot \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \partial_t u_m^* | 0 \rangle = \partial_t u_n^* \partial_t u_m^* \langle 0 | \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle = 0$$

These results follow from

$$\hat{a}_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle = 0$$

(the definition of the vacuum state) and

$$\langle 0 | \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle = \langle 0 | 1_{n,\boldsymbol{k}_{\perp,n}}, 1_{m,\boldsymbol{k}_{\perp,m}} \rangle = 0$$

(the two states are orthogonal), or equivalently,

$$\langle 0 | \hat{a}^\dagger_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} | 0 \rangle = \left(\hat{a}_{m,\boldsymbol{k}_{\perp,m}} \hat{a}_{n,\boldsymbol{k}_{\perp,n}} | 0 \rangle \right)^\dagger | 0 \rangle = 0^\dagger \cdot | 0 \rangle = 0$$

From

$$\left[\hat{a}_{n,\boldsymbol{k}_{\perp,n}}, \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \right] = \hat{a}_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} - \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \hat{a}_{n,\boldsymbol{k}_{\perp,n}} = \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm}$$

we obtain

$$\hat{a}_{n,\boldsymbol{k}_{\perp,n}} \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} = \delta^{(2)} \left(\boldsymbol{k}_{\perp,n} - \boldsymbol{k}_{\perp,m} \right) \delta_{nm} + \hat{a}^\dagger_{m,\boldsymbol{k}_{\perp,m}} \hat{a}_{n,\boldsymbol{k}_{\perp,n}}$$

We previously obtained

$$\langle 0 | \hat{T}_{tt} | 0 \rangle = \sum_n \int d^2k_\perp \left[\partial_t u_n \partial_t u_n^* - \frac{1}{2} \hat{g}_{tt} \left(\hat{g}^{\mu\nu} \partial_\mu u_n \partial_\nu u_n^* \right) \right]$$

where $\int d^2k_{\perp,n}$ is denoted by $\int d^2k_\perp$. We also have

$$u_n = N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L} x \right)$$

$$e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} = e^{i(k_y y + k_z z)}$$

$$\hat{g}_{\mu\nu} = \begin{pmatrix} C^{-2}(\Omega) & -\frac{A}{r^3}(\Omega - \omega_d) & 0 & 0 \\ -\frac{A}{r^3}(\Omega - \omega_d) & -\frac{A}{r^4} & 0 & 0 \\ 0 & 0 & -\frac{r^2}{\Delta} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

$$\hat{g}^{\mu\nu} = \begin{pmatrix} \frac{A}{\Delta r^2} & \frac{A}{\Delta r}(\omega_d - \Omega) & 0 & 0 \\ \frac{A}{\Delta r}(\omega_d - \Omega) & -\frac{r^2}{\Delta} C^{-2}(\Omega) & 0 & 0 \\ 0 & 0 & -\frac{\Delta}{r^2} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

$$C(\Omega) = \left\{ \frac{\Delta r^2}{A} \left[1 - \frac{A^2}{\Delta r^4} (\Omega - \omega_d)^2 \right] \right\}^{-1/2},$$

$$N_n^2 = \frac{1}{(2\pi)^2 L \omega_n C^2(\Omega)}$$

$$\omega_n = \frac{r}{\sqrt{\Delta}} C^{-2}(\Omega) \sqrt{\left(\frac{n\pi}{L}\right)^2 + \frac{\Delta}{r^2} C^2(\Omega) \left(\frac{\Delta}{r^2} k_y^2 + k_z^2\right)}$$

$$\beta_n = \frac{b_n A}{r^3} (\omega_d - \Omega)$$

$$b_n = \omega_n C^2(\Omega)$$

Thus,

$$\begin{aligned}
\partial_t u_n \partial_t u_n^* &= \partial_t N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \partial_t N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\
&= (-i\omega_n)\, i\omega_n N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\
&= \omega_n^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right)
\end{aligned}$$

$$\begin{aligned}
\partial_t u_n \partial_x u_n^* &= \partial_t N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \partial_x N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\
&= \left(-i\omega_n\right) N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} \left[i\beta_n e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \right. \\
&\quad \left. + e^{i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) \right] \\
&= \left(-i\omega_n\right) i\beta_n N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) + \Big(\\
&\quad - i\omega_n \Big) N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) \\
&= \omega_n \beta_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \left(-i\omega_n\right) \frac{n\pi}{L} N_n^2 \sin\left(\frac{n\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right)
\end{aligned}$$

$$\begin{aligned}
\partial_x u_n \partial_t u_n^* &= \partial_x N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \partial_t N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\
&= N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} \left[\left(-i\beta_n\right) e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) + e^{-i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) \right] i\omega_n N_n e^{i\omega_n t} \\
&\quad e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\
&= \left(-i\beta_n\right) i\omega_n N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) + i\omega_n \\
&\quad N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\
&= \beta_n \omega_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + i\omega_n \frac{n\pi}{L} N_n^2 \cos\left(\frac{n\pi}{L}x\right) \sin\left(\frac{n\pi}{L}x\right)
\end{aligned}$$

$$\begin{aligned}
\partial_x u_n \partial_x u_n^* &= \partial_x N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \partial_x N_n e^{i\omega_n t} e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\
&= N_n e^{-i\omega_n t} e^{i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} \left[\left(-i\beta_n\right) e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) + e^{-i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) \right] N_n e^{i\omega_n t} \\
&\quad e^{-i\boldsymbol{k}_\perp \cdot \boldsymbol{r}_\perp} \left[i\beta_n e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) + e^{i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) \right] \\
&= N_n^2 \Bigg[\left(-i\beta_n\right) e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) i\beta_n e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) + \left(-i\beta_n\right) e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) e^{i\beta_n x} \\
&\quad \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) + e^{-i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) i\beta_n e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) + e^{-i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}\right. \\
&\quad \left. x\right) e^{i\beta_n x} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi}{L}x\right) \Bigg] \\
&= N_n^2 \Bigg[\beta_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \left(-i\beta_n\right) \frac{n\pi}{L} \sin\left(\frac{n\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) + i\beta_n \frac{n\pi}{L} \cos\left(\frac{n\pi}{L}x\right) \sin\left(\frac{n\pi}{L}\right.
\end{aligned}$$

$$x\Big) + \left(\frac{n\pi}{L}\right)^2 \cos^2\left(\frac{n\pi}{L}x\right)\Big]$$
$$= N_n^2\Big[\beta_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \left(\frac{n\pi}{L}\right)^2 \cos^2\left(\frac{n\pi}{L}x\right)\Big]$$

$$\begin{aligned}\partial_y u_n \partial_y u_n^* &= \partial_y N_n e^{-i\omega_n t} e^{i(k_y y + k_z z)} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \partial_y N_n e^{i\omega_n t} e^{-i(k_y y + k_z z)} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\ &= ik_y\left(-ik_y\right) N_n e^{-i\omega_n t} e^{i(k_y y + k_z z)} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i(k_y y + k_z z)} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\ &= k_y^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right)\end{aligned}$$

$$\begin{aligned}\partial_z u_n \partial_z u_n^* &= \partial_z N_n e^{-i\omega_n t} e^{i(k_y y + k_z z)} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \partial_z N_n e^{i\omega_n t} e^{-i(k_y y + k_z z)} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\ &= ik_z\left(-ik_z\right) N_n e^{-i\omega_n t} e^{i(k_y y + k_z z)} e^{-i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) N_n e^{i\omega_n t} e^{-i(k_y y + k_z z)} e^{i\beta_n x} \sin\left(\frac{n\pi}{L}x\right) \\ &= k_z^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right)\end{aligned}$$

$$\begin{aligned}&\frac{1}{2}\hat{g}_{tt}\left(\hat{g}^{\mu\nu}\partial_\mu u_n \partial_\nu u_n^*\right) = \frac{1}{2}\hat{g}_{tt}\Big(\hat{g}^{tt}\partial_t u_n \partial_t u_n^* + \hat{g}^{tx}\partial_t u_n \partial_x u_n^* + \hat{g}^{xt}\partial_x u_n \partial_t u_n^* \\ &\quad + \hat{g}^{xx}\partial_x u_n \partial_x u_n^* + \hat{g}^{yy}\partial_y u_n \partial_y u_n^* + \hat{g}^{zz}\partial_z u_n \partial_z u_n^*\Big) \\ &= \frac{1}{2}\hat{g}_{tt}\Big\{\hat{g}^{tt}\omega_n^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{tx}\Big[\omega_n\beta_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \left(-i\omega_n\right)\frac{n\pi}{L}N_n^2 \sin\Big(\frac{n\pi}{L} \\ &\quad x\Big)\cos\left(\frac{n\pi}{L}x\right)\Big] + \hat{g}^{xt}\Big[\beta_n\omega_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + i\omega_n\frac{n\pi}{L}N_n^2 \cos\left(\frac{n\pi}{L}x\right)\sin\left(\frac{n\pi}{L}x\right)\Big] + \hat{g}^{xx} \\ &\quad N_n^2\Big[\beta_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \left(\frac{n\pi}{L}\right)^2 \cos^2\left(\frac{n\pi}{L}x\right)\Big] + \hat{g}^{yy}k_y^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{zz}k_z^2 N_n^2 \sin^2 \\ &\quad \left(\frac{n\pi}{L}x\right)\Big\} \\ &= \frac{1}{2}\hat{g}_{tt}\Big\{\hat{g}^{tt}\omega_n^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{tx}\omega_n\beta_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{xt}\beta_n\omega_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) \\ &\quad + \hat{g}^{xx}N_n^2\Big[\beta_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \left(\frac{n\pi}{L}\right)^2 \cos^2\left(\frac{n\pi}{L}x\right)\Big] + \hat{g}^{yy}k_y^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{zz}k_z^2 N_n^2 \\ &\quad \sin^2\left(\frac{n\pi}{L}x\right)\Big\} \\ &= \frac{1}{2}\hat{g}_{tt}\Big\{\hat{g}^{tt}\omega_n^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + 2\hat{g}^{tx}\omega_n\beta_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{xx}N_n^2\Big[\beta_n^2 \sin^2\left(\frac{n\pi}{L}x\right) \\ &\quad + \left(\frac{n\pi}{L}\right)^2 \cos^2\left(\frac{n\pi}{L}x\right)\Big] + \hat{g}^{yy}k_y^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{zz}k_z^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right)\Big\}\end{aligned}$$

Hence,

$$\begin{aligned}&\partial_t u_n \partial_t u_n^* - \frac{1}{2}\hat{g}_{tt}\left(\hat{g}^{\mu\nu}\partial_\mu u_n \partial_\nu u_n^*\right) = \omega_n^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) - \frac{1}{2}\hat{g}_{tt}\Big\{\hat{g}^{tt}\omega_n^2 N_n^2 \sin^2\Big(\frac{n\pi}{L} \\ &\quad x\Big) + 2\hat{g}^{tx}\omega_n\beta_n N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{xx}N_n^2\Big[\beta_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \left(\frac{n\pi}{L}\right)^2 \cos^2\left(\frac{n\pi}{L}x\right)\Big] + \hat{g}^{yy} \\ &\quad k_y^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right) + \hat{g}^{zz}k_z^2 N_n^2 \sin^2\left(\frac{n\pi}{L}x\right)\Big\}\end{aligned}$$

Write $k_n = \frac{n\pi}{L}$, $S_x = \sin^2(k_n x)$, and $C_x = \cos^2(k_n x)$. Then

$$\begin{aligned}&\partial_t u_n \partial_t u_n^* - \frac{1}{2}\hat{g}_{tt}\left(\hat{g}^{\mu\nu}\partial_\mu u_n \partial_\nu u_n^*\right) = \omega_n^2 N_n^2 S_x - \frac{1}{2}\hat{g}_{tt}\Big[\hat{g}^{tt}\omega_n^2 N_n^2 S_x + 2\hat{g}^{tx}\omega_n\beta_n N_n^2 \\ &\quad S_x + \hat{g}^{xx}N_n^2\left(\beta_n^2 S_x + k_n^2 C_x\right) + \hat{g}^{yy}k_y^2 N_n^2 S_x + \hat{g}^{zz}k_z^2 N_n^2 S_x\Big]\end{aligned}$$

$$
\begin{aligned}
&= \omega_n^2 N_n^2 S_x - \frac{1}{2}\hat{g}_{tt}\left[\left(\hat{g}^{tt}\omega_n^2 + 2\hat{g}^{tx}\omega_n\beta_n + \hat{g}^{xx}\beta_n^2\right)N_n^2 S_x + \hat{g}^{xx}N_n^2 k_n^2 C_x + \hat{g}^{yy}k_y^2 N_n^2 S_x + \hat{g}^{zz}k_z^2 N_n^2 S_x\right] \\
&= \omega_n^2 N_n^2 S_x - \frac{1}{2}\hat{g}_{tt}\left[\omega_n^2 C^2\left(\Omega\right)N_n^2 S_x + \hat{g}^{xx}N_n^2 k_n^2 C_x + \hat{g}^{yy}k_y^2 N_n^2 S_x + \hat{g}^{zz}k_z^2 N_n^2 S_x\right] \\
&= \omega_n^2 N_n^2 S_x - \frac{1}{2}C^{-2}\left(\Omega\right)\left\{\omega_n^2 C^2\left(\Omega\right)N_n^2 S_x + \left[-\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\right]N_n^2 k_n^2 C_x + \left(-\frac{\Delta}{r^2}\right)\right. \\
&\quad \left. k_y^2 N_n^2 S_x + \left(-1\right)k_z^2 N_n^2 S_x\right\} \\
&= \omega_n^2 N_n^2 S_x - \frac{1}{2}N_n^2\left\{\omega_n^2 S_x - \frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2 C_x - C^{-2}\left(\Omega\right)\frac{\Delta}{r^2}k_y^2 S_x - C^{-2}\left(\Omega\right)k_z^2 S_x\right\} \\
&= N_n^2\left[\frac{1}{2}\omega_n^2 S_x + \frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2 C_x + \frac{1}{2}C^{-2}\left(\Omega\right)\left(\frac{\Delta}{r^2}k_y^2 + k_z^2\right)S_x\right] \\
&= N_n^2\left[\frac{1}{2}\omega_n^2 S_x + \frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2 C_x + \frac{1}{2}\left[\omega_n^2 - \frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2\right]S_x\right] \\
&= N_n^2\left[\omega_n^2 S_x + \frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2 C_x - \frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2 S_x\right] \\
&= N_n^2\left[\omega_n^2 S_x + \frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2\left(C_x - S_x\right)\right] \\
&= N_n^2\left[\omega_n^2 S_x + \frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2\cos\left(2k_n x\right)\right] \\
&= N_n^2\left[\omega_n^2\sin^2\left(\frac{n\pi}{L}x\right) + \frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2\cos\left(\frac{2n\pi}{L}x\right)\right]
\end{aligned}
$$

where

$$
\begin{aligned}
\hat{g}^{tt}\omega_n^2 + 2\hat{g}^{tx}\omega_n\beta_n + \hat{g}^{xx}\beta_n^2 &= \frac{A}{\Delta r^2}\omega_n^2 + 2\frac{A}{\Delta r}\left(\omega_d - \Omega\right)\omega_n \cdot \frac{\omega_n C^2\left(\Omega\right)A}{r^3}\left(\omega_d - \Omega\right) + \left[-\frac{r^2}{\Delta}\right. \\
&\quad \left. C^{-2}\left(\Omega\right)\right]\left[\frac{\omega_n C^2\left(\Omega\right)A}{r^3}\left(\omega_d - \Omega\right)\right]^2 \\
&= \frac{A}{\Delta r^2}\omega_n^2 + 2\frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\omega_n^2 C^2\left(\Omega\right) - \frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\frac{\omega_n^2 C^4\left(\Omega\right)A^2}{r^6}\left(\omega_d - \Omega\right)^2 \\
&= \frac{A}{\Delta r^2}\omega_n^2 + 2\frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\omega_n^2 C^2\left(\Omega\right) - \frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\omega_n^2 C^2\left(\Omega\right) \\
&= \frac{A}{\Delta r^2}\omega_n^2 + \frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\omega_n^2 C^2\left(\Omega\right)
\end{aligned}
$$

and

$$
\begin{aligned}
\frac{A}{\Delta r^2}\omega_n^2 + \frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\omega_n^2 C^2\left(\Omega\right) &= \left[\frac{A}{\Delta r^2}C^{-2}\left(\Omega\right) + \frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\right]\omega_n^2 C^2\left(\Omega\right) \\
&= \left\{\frac{A}{\Delta r^2}\cdot\frac{\Delta r^2}{A}\left[1 - \frac{A^2}{\Delta r^4}\left(\Omega - \omega_d\right)^2\right] + \frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\right\}\omega_n^2 C^2\left(\Omega\right) \\
&= \left[1 - \frac{A^2}{\Delta r^4}\left(\Omega - \omega_d\right)^2 + \frac{A^2}{\Delta r^4}\left(\omega_d - \Omega\right)^2\right]\omega_n^2 C^2\left(\Omega\right) \\
&= \omega_n^2 C^2\left(\Omega\right)
\end{aligned}
$$

This gives

$$
\hat{g}^{tt}\omega_n^2 + 2\hat{g}^{tx}\omega_n\beta_n + \hat{g}^{xx}\beta_n^2 = \omega_n^2 C^2\left(\Omega\right)
$$

Also,

$$
\begin{aligned}
\omega_n^2 &= \frac{r^2}{\Delta}C^{-4}\left(\Omega\right)\left[k_n^2 + \frac{\Delta}{r^2}C^2\left(\Omega\right)\left(\frac{\Delta}{r^2}k_y^2 + k_z^2\right)\right] \\
&= \frac{r^2}{\Delta}C^{-4}\left(\Omega\right)k_n^2 + C^{-2}\left(\Omega\right)\left(\frac{\Delta}{r^2}k_y^2 + k_z^2\right)
\end{aligned}
$$

which gives

$$C^{-2}(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)=\omega_n^2-\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2$$

Finally,

$$C_x-S_x=\cos^2(k_nx)-\sin^2(k_nx)=\cos(2k_nx)$$

by the double-angle identity $\cos 2\alpha=\cos^2\alpha-\sin^2\alpha$.

Alternatively,

$$\begin{aligned}
\partial_t u_n\partial_t u_n^*-\frac{1}{2}\hat{g}_{tt}\left(\hat{g}^{\mu\nu}\partial_\mu u_n\partial_\nu u_n^*\right)&=N_n^2\left[\omega_n^2\sin^2\left(\frac{n\pi}{L}x\right)+\frac{1}{2}\frac{r^2}{\Delta}C^{-4}(\Omega)\left(\frac{n\pi}{L}\right)^2\cos\left(\frac{2n\pi}{L}x\right)\right]\\
&=N_n^2\left\{\omega_n^2\sin^2(k_nx)+\frac{1}{2}\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2\left[\cos^2(k_nx)-\sin^2(k_nx)\right]\right\}\\
&=N_n^2\left\{\left[\omega_n^2-\frac{1}{2}\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2\right]\sin^2(k_nx)+\frac{1}{2}\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2\cos^2(k_nx)\right\}\\
&=\frac{1}{2}N_n^2\left\{\left[2\omega_n^2-\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2\right]\sin^2(k_nx)+\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2\cos^2(k_nx)\right\}\\
&=\frac{1}{2}N_n^2\left\{\left[\omega_n^2+\frac{\omega_n}{b_n}\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\right]\sin^2(k_nx)+\frac{\omega_n^2r^2}{b_n^2\Delta}k_n^2\cos^2(k_nx)\right\}\\
&=\frac{1}{2}N_n^2\left\{\mathcal{F}_n\sin^2(k_nx)+\mathcal{G}_n\cos^2(k_nx)\right\}
\end{aligned}$$

where we define

$$\mathcal{F}_n=\omega_n^2+\frac{\omega_n}{b_n}\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)$$

$$\mathcal{G}_n=\frac{\omega_n^2r^2}{b_n^2\Delta}k_n^2$$

We also have

$$\omega_n=\frac{r}{\sqrt{\Delta}}C^{-2}(\Omega)\sqrt{k_n^2+\frac{\Delta}{r^2}C^2(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)}$$

$$\omega_n^2=\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2+C^{-2}(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)$$

$$\beta_n=\frac{b_nA}{r^3}(\omega_d-\Omega)$$

$$b_n=\omega_nC^2(\Omega)$$

$$k_n=\frac{n\pi}{L}$$

$$\begin{aligned}
2\omega_n^2-\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2&=\omega_n^2+\omega_n^2-\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2\\
&=\omega_n^2+\left[\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2+C^{-2}(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\right]-\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2\\
&=\omega_n^2+C^{-2}(\Omega)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\\
&=\omega_n^2+\frac{\omega_n}{\omega_nC^2(\Omega)}\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\\
&=\omega_n^2+\frac{\omega_n}{b_n}\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)
\end{aligned}$$

$$\begin{aligned}
\frac{r^2}{\Delta}C^{-4}(\Omega)\,k_n^2&=\frac{\omega_n^2r^2}{\omega_n^2C^4(\Omega)\,\Delta}k_n^2\\
&=\frac{\omega_n^2r^2}{b_n^2\Delta}k_n^2
\end{aligned}$$

We obtain

$$\begin{aligned}
\langle\epsilon_{\text{vac}}\rangle &= C^2\left(\Omega\right)\frac{1}{L\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz}\int_0^L dx\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz\langle 0|T_{tt}|0\rangle \\
&= C^2\left(\Omega\right)\frac{1}{L\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz}\int_0^L dx\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz\sum_n\int d^2k_\perp\left[\partial_t u_n\partial_t u_n^* - \frac{1}{2}\hat{g}_{tt}\left(\hat{g}^{\mu\nu}\partial_\mu u_n\partial_\nu u_n^*\right)\right] \\
&= C^2\left(\Omega\right)\frac{1}{L\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz}\int_0^L dx\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz\sum_n\int d^2k_\perp\left\{N_n^2\left[\omega_n^2\sin^2\left(\frac{n\pi}{L}x\right)+\frac{1}{2}\right.\right. \\
&\quad \left.\left.\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2\cos\left(\frac{2n\pi}{L}x\right)\right]\right\} \\
&= \frac{C^2\left(\Omega\right)}{L}\int_0^L dx\sum_n\int d^2k_\perp N_n^2\left[\omega_n^2\sin^2\left(\frac{n\pi}{L}x\right)+\frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2\cos\left(\frac{2n\pi}{L}x\right)\right] \\
&= \frac{C^2\left(\Omega\right)}{L}\sum_n\int d^2k_\perp N_n^2\left[\omega_n^2\int_0^L\sin^2\left(\frac{n\pi}{L}x\right)dx+\frac{1}{2}\frac{r^2}{\Delta}C^{-4}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2\int_0^L\cos\left(\frac{2n\pi}{L}x\right)dx\right] \\
&= \frac{C^2\left(\Omega\right)}{L}\sum_n\int d^2k_\perp N_n^2\omega_n^2\frac{L}{2} \\
&= \frac{C^2\left(\Omega\right)}{2}\sum_n\int d^2k_\perp N_n^2\omega_n^2
\end{aligned}$$

Here, $\langle 0|T_{tt}|0\rangle$ is independent of y,z, so the integral $\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}dydz$ can be factored out. We have $\int_0^L\sin^2\left(\frac{n\pi}{L}x\right)dx = \int_0^L\frac{1-\cos\left(2\frac{n\pi}{L}x\right)}{2}dx = \frac{L}{2}$ and $\int_0^L\cos\left(\frac{2n\pi}{L}x\right)dx = \frac{L}{2n\pi}\sin\left(\frac{2n\pi}{L}x\right)\Big|_0^L = \frac{L}{2n\pi}\left[\sin(2n\pi)-0\right] = 0$ (since $\sin 2n\pi = 0$). Also, $N_n^2 = \frac{1}{(2\pi)^2 L\omega_n C^2(\Omega)}$.

$$\begin{aligned}
\langle\epsilon_{\text{vac}}\rangle &= \frac{C^2\left(\Omega\right)}{2}\sum_n\int d^2k_\perp N_n^2\omega_n^2 \\
&= \frac{C^2\left(\Omega\right)}{2}\sum_n\int d^2k_\perp\frac{1}{\left(2\pi\right)^2 L\omega_n C^2\left(\Omega\right)}\omega_n^2 \\
&= \frac{1}{2\left(2\pi\right)^2 L}\sum_n\int d^2k_\perp\omega_n \\
&= \frac{1}{2\left(2\pi\right)^2 L}\sum_n\int d^2k_\perp\left[\frac{r}{\sqrt{\Delta}}C^{-2}\left(\Omega\right)\sqrt{\left(\frac{n\pi}{L}\right)^2+\frac{\Delta}{r^2}C^2\left(\Omega\right)\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)}\right] \\
&= \frac{1}{2\left(2\pi\right)^2 L}\sum_n\int d^2k_\perp\left\{\frac{r}{\sqrt{\Delta}}C^{-2}\left(\Omega\right)\sqrt{\frac{\Delta}{r^2}C^2\left(\Omega\right)\left[\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2+\left(\frac{\Delta}{r^2}k_y^2+k_z^2\right)\right]}\right\} \\
&= \frac{1}{2\left(2\pi\right)^2 L}\sum_n\int d^2k_\perp\left[\frac{r}{\sqrt{\Delta}}C^{-2}\left(\Omega\right)\cdot\frac{\sqrt{\Delta}}{r}C\left(\Omega\right)\sqrt{\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2+\frac{\Delta}{r^2}k_y^2+k_z^2}\right] \\
&= \frac{1}{2\left(2\pi\right)^2 L}\sum_n\int d^2k_\perp\left[C^{-1}\left(\Omega\right)\sqrt{\frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2+\frac{\Delta}{r^2}k_y^2+k_z^2}\right]
\end{aligned}$$

Define

$$s = \frac{r^2}{\Delta}C^{-2}\left(\Omega\right)\left(\frac{n\pi}{L}\right)^2$$

$$\tilde{k}_y = \frac{\sqrt{\Delta}}{r}k_y$$

Then

$$\begin{aligned}\sum_n \int d^2k_\perp \omega_n &= \sum_n \int d^2k_\perp \left[C^{-1}(\Omega)\sqrt{\frac{r^2}{\Delta}C^{-2}(\Omega)\left(\frac{n\pi}{L}\right)^2+\frac{\Delta}{r^2}k_y^2+k_z^2}\right] \\ &= C^{-1}(\Omega)\sum_n \int d^2k_\perp \sqrt{s+\tilde{k}_y^2+k_z^2} \\ &= \frac{r}{C(\Omega)\sqrt{\Delta}}\sum_n \int d^2\tilde{k}_\perp \left(s+\tilde{k}_y^2+k_z^2\right)^{1/2}\end{aligned}$$

where $k_y=\frac{r}{\sqrt{\Delta}}\tilde{k}_y$, $dk_y=d\left(\frac{r}{\sqrt{\Delta}}\tilde{k}_y\right)$, and $d^2k_\perp=dk_ydk_z=d\left(\frac{r}{\sqrt{\Delta}}\tilde{k}_y\right)dk_z=\frac{r}{\sqrt{\Delta}}d\tilde{k}_ydk_z$; also, $d^2\tilde{k}_\perp=d\tilde{k}_ydk_z$. Apply the Schwinger proper-time method to evaluate the momentum-space integral:

$$\begin{aligned}\int d^2\tilde{k}_\perp \left(s+\tilde{k}_y^2+k_z^2\right)^{1/2} &= \int d^2\tilde{k}_\perp \frac{1}{\Gamma(-1/2)}\int_0^\infty dt t^{-\frac{1}{2}-1}e^{-t\left(s+\tilde{k}_y^2+k_z^2\right)} \\ &= \int d^2\tilde{k}_\perp \left[-\frac{1}{2\sqrt{\pi}}\int_0^\infty dt t^{-\frac{3}{2}}e^{-t\left(s+\tilde{k}_y^2+k_z^2\right)}\right] \\ &= -\frac{1}{2\sqrt{\pi}}\int_0^\infty dt t^{-\frac{3}{2}}\int_{-\infty}^\infty e^{-t\tilde{k}_y^2}d\tilde{k}_y\int_{-\infty}^\infty e^{-tk_z^2}dk_z e^{-ts} \\ &= -\frac{1}{2\sqrt{\pi}}\int_0^\infty dt t^{-\frac{3}{2}}\frac{\pi}{t}e^{-ts} \\ &= -\frac{\sqrt{\pi}}{2}\int_0^\infty dt t^{-\frac{3}{2}-1}e^{-ts} \\ &= -\frac{\sqrt{\pi}}{2}s^{3/2}\Gamma\left(-\frac{3}{2}\right) \\ &= -\frac{\sqrt{\pi}}{2}s^{3/2}\frac{4\sqrt{\pi}}{3} \\ &= -\frac{2\pi}{3}s^{3/2}\end{aligned}$$

Here, after analytic continuation, $A^{-\alpha}=\frac{1}{\Gamma(\alpha)}\int_0^\infty dt t^{\alpha-1}e^{-tA}$; $\frac{1}{\Gamma(-1/2)}=-\frac{1}{2\sqrt{\pi}}$; $\int_{-\infty}^\infty e^{-t\tilde{k}_y^2}d\tilde{k}_y=\int_{-\infty}^\infty e^{-tk_z^2}dk_z=\sqrt{\frac{\pi}{t}}$; $\int_0^\infty t^{z-1}e^{-ts}dt=s^{-z}\Gamma(z)$; and $\Gamma\left(-\frac{3}{2}\right)=\frac{4\sqrt{\pi}}{3}$. Hence,

$$\begin{aligned}\sum_n \int d^2k_\perp \omega_n &= \frac{r}{C(\Omega)\sqrt{\Delta}}\sum_n \int d^2\tilde{k}_\perp \left(s+\tilde{k}_y^2+k_z^2\right)^{1/2} \\ &= \frac{r}{C(\Omega)\sqrt{\Delta}}\sum_n \left(-\frac{2\pi}{3}s^{3/2}\right) \\ &= \frac{r}{C(\Omega)\sqrt{\Delta}}\sum_n \left\{-\frac{2\pi}{3}\left[\frac{r^2}{\Delta}C^{-2}(\Omega)\left(\frac{n\pi}{L}\right)^2\right]^{3/2}\right\} \\ &= \frac{r^4}{C^4(\Omega)\Delta^2}\sum_n \left[-\frac{2\pi}{3}\left(\frac{n\pi}{L}\right)^3\right] \\ &= -\frac{2\pi^4}{3L^3}\frac{r^4}{C^4(\Omega)\Delta^2}\sum_n n^3\end{aligned}$$

Applying Riemann ζ-function regularization, $\sum_{n=1}^\infty n^3\to\zeta(-3)=\frac{1}{120}$. We obtain

$$\begin{aligned}\sum_n \int d^2k_\perp \omega_n &= -\frac{2\pi^4}{3L^3}\frac{r^4}{C^4(\Omega)\Delta^2}\cdot\frac{1}{120} \\ &= -\frac{\pi^4}{180L^3}\frac{r^4}{C^4(\Omega)\Delta^2}\end{aligned}$$

and consequently

$$\begin{aligned}\langle\epsilon_{\rm vac}\rangle &= \frac{1}{2(2\pi)^2 L}\sum_n \int d^2k_\perp \omega_n \\ &= \frac{1}{8\pi^2 L}\left(-\frac{\pi^4}{180L^3}\frac{r^4}{C^4(\Omega)\Delta^2}\right) \\ &= -\frac{\pi^2}{1440L^4}\frac{r^4}{C^4(\Omega)\Delta^2}\end{aligned}$$

The proper length of the Casimir cavity is obtained as follows. For $ds^2 = \hat{g}_{tt}d\tilde{t}^2 + \left(\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right)dx^2 + \hat{g}_{yy}dy^2 + \hat{g}_{zz}dz^2$ with $d\tilde{t} = dy = dz = 0$, a spacelike interval is negative under the signature $(+,-,-,-)$, so the absolute value must be taken under the square root: $ds = \sqrt{\left|\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right|}dx$. Thus,

$$L_p = \int_0^L ds = \int_0^L \sqrt{\left|\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right|}dx$$

With the metric approximated as constant,

$$\begin{aligned}
L_p &= \sqrt{\left|\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right|}\int_0^L dx \\
&= L\sqrt{\left|\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right|} \\
&= L\sqrt{\left|-\frac{\Delta}{r^2}C^2(\Omega)\right|} \\
&= L\sqrt{\frac{\Delta}{r^2}C^2(\Omega)} \\
&= L\frac{\sqrt{\Delta}}{r}C(\Omega)
\end{aligned}$$

where

$$\begin{aligned}
\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}} &= -\frac{A}{r^4} - \frac{\left[-\frac{A}{r^3}(\Omega-\omega_d)\right]^2}{C^{-2}(\Omega)} \\
&= -\frac{A}{r^4} - \frac{\frac{A^2}{r^6}(\Omega-\omega_d)^2}{C^{-2}(\Omega)} \\
&= -\frac{A}{r^4} - \frac{\frac{A^2}{r^6}(\Omega-\omega_d)^2}{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]} \\
&= -\frac{A}{r^4} - \frac{A}{r^4}\frac{\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2} \\
&= \frac{A}{r^4}\left[-1 + \frac{-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2}\right] \\
&= \frac{A}{r^4}\left[-1 + \left(1 - \frac{1}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2}\right)\right] \\
&= -\frac{A}{r^4}\frac{1}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2} \\
&= -\frac{\Delta}{r^2}\frac{1}{\frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]} \\
&= -\frac{\Delta}{r^2}\frac{1}{C^{-2}(\Omega)} \\
&= -\frac{\Delta}{r^2}C^2(\Omega)
\end{aligned}$$

Here,

$$\begin{aligned}
\frac{-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2} &= \frac{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2} - \frac{1}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2} \\
&= 1 - \frac{1}{1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2}
\end{aligned}$$

$$C^{-2}(\Omega) = \frac{\Delta r^2}{A}\left[1-\frac{A^2}{\Delta r^4}(\Omega-\omega_d)^2\right]$$

We obtain

$$L = L_p\frac{r}{C(\Omega)\sqrt{\Delta}}$$

$$L^4 = L_p^4 \frac{r^4}{C^4(\Omega)\Delta^2}$$

The Casimir energy per unit volume is therefore

$$\begin{aligned}
\langle \epsilon_{\text{vac}} \rangle &= -\frac{\pi^2}{1440 L^4} \frac{r^4}{C^4(\Omega)\Delta^2} \\
&= -\frac{\pi^2}{1440 \left[L_p^4 \frac{r^4}{C^4(\Omega)\Delta^2} \right]} \frac{r^4}{C^4(\Omega)\Delta^2} \\
&= -\frac{\pi^2}{1440 L_p^4} \frac{C^4(\Omega)\Delta^2}{r^4} \frac{r^4}{C^4(\Omega)\Delta^2} \\
&= -\frac{\pi^2}{1440 L_p^4}.
\end{aligned}$$

This is precisely the expression for the Casimir energy density in flat spacetime (Minkowski spacetime).

VI. Further Perspectives: Geometry and Boundary Conditions in the Constant-Metric Approximation

Once the metric is approximated as constant, the spacetime is in fact Minkowski spacetime. A linear coordinate transformation with constant coefficients brings the line element into the standard Minkowski form. Let $\bar{g}_{\mu\nu}$ denote the metric components after this approximation. Completing the square gives

$$ds^2 = \bar{g}_{tt} \left(dt + \frac{\bar{g}_{tx}}{\bar{g}_{tt}} dx \right)^2 + \left(\bar{g}_{xx} - \frac{\bar{g}_{tx}^2}{\bar{g}_{tt}} \right) dx^2 + \bar{g}_{yy} dy^2 + \bar{g}_{zz} dz^2,$$

where all metric coefficients on the right-hand side are fixed constants. Set

$$\begin{aligned}
dT &= \sqrt{\bar{g}_{tt}} \left(dt + \frac{\bar{g}_{tx}}{\bar{g}_{tt}} dx \right), \\
dX &= \sqrt{-\left(\bar{g}_{xx} - \frac{\bar{g}_{tx}^2}{\bar{g}_{tt}} \right)} dx, \\
dY &= \sqrt{-\bar{g}_{yy}} dy, \\
dZ &= \sqrt{-\bar{g}_{zz}} dz,
\end{aligned}$$

or apply the affine transformation

$$\begin{aligned}
T - T_0 &= \sqrt{\bar{g}_{tt}} \left[(t - t_0) + \frac{\bar{g}_{tx}}{\bar{g}_{tt}} (x - x_0) \right], \\
X - X_0 &= \sqrt{-\left(\bar{g}_{xx} - \frac{\bar{g}_{tx}^2}{\bar{g}_{tt}} \right)} (x - x_0), \\
Y - Y_0 &= \sqrt{-\bar{g}_{yy}} (y - y_0), \\
Z - Z_0 &= \sqrt{-\bar{g}_{zz}} (z - z_0),
\end{aligned}$$

Here, all radicands are greater than 0 when the worldline of the cavity center is timelike and $r_0 > r_+$, $\theta_0 = \pi/2$. All coordinate values with subscript 0 are constants and can be set to 0 by suitable choices of the respective coordinate origins. We then obtain

$$ds^2 = dT^2 - dX^2 - dY^2 - dZ^2.$$

In this coordinate system, the Dirichlet boundary conditions become

$$\Psi(T, X = 0, Y, Z) = \Psi\left(T, X = \frac{\sqrt{\Delta}}{r} C(\Omega) L, Y, Z \right) = 0,$$

When

$$x = L$$

we have

$$X = \sqrt{-\left(\hat{g}_{xx} - \frac{\hat{g}_{tx}^2}{\hat{g}_{tt}}\right)} \cdot L$$
$$= \frac{\sqrt{\Delta}}{r} C(\Omega) L.$$

This also explains why, after transforming the line element into the standard Minkowski form, the positions of the two parallel plates are no longer both expressed in the same form as $X = 0$ and $X = L$.

From the viewpoint of spacetime curvature, approximating the metric components as constants gives

$$\partial_\lambda \bar{g}_{\mu\nu} = 0,$$

$$\bar{\Gamma}^\sigma_{\mu\nu} = \frac{1}{2}\bar{g}^{\sigma\rho}\left(\partial_\mu \bar{g}_{\nu\rho} + \partial_\nu \bar{g}_{\rho\mu} - \partial_\rho \bar{g}_{\mu\nu}\right) = 0,$$

$$\bar{R}^\rho_{\sigma\mu\nu} = \partial_\mu \bar{\Gamma}^\rho_{\nu\sigma} - \partial_\nu \bar{\Gamma}^\rho_{\mu\sigma} + \bar{\Gamma}^\rho_{\mu\lambda}\bar{\Gamma}^\lambda_{\nu\sigma} - \bar{\Gamma}^\rho_{\nu\lambda}\bar{\Gamma}^\lambda_{\mu\sigma} = 0,$$

Thus, every component of the Riemann curvature tensor is 0. This is precisely what it means for spacetime to be flat, and its physical properties are naturally those of flat spacetime. Unless otherwise necessary, the corresponding physical problem can therefore be studied entirely in flat spacetime.

For parallel plates in flat spacetime undergoing circular motion about an external center, the Casimir energy need not be the same as when the plates are at rest in an inertial frame, because their motion is noninertial. In the present analysis and in the original paper under discussion, however, this motion is effectively approximated as inertial. This can be understood in two ways for parallel plates in circular motion. First, in a coordinate system at rest relative to the plates, the metric depends on position. Second, in standard inertial Minkowski coordinates, the boundary conditions depend on time. In the present analysis, adopting the constant-metric approximation allows a transformation to the standard Minkowski form of the metric, while time-independent boundary conditions similar to those for plates at rest in an inertial frame are still imposed. This clearly describes the physical situation of stationary parallel plates, rather than that of parallel plates in circular motion.